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14th Edition

CLSI M02™

Performance Standards for Antimicrobial Disk Susceptibility Tests

CLSI M02 covers the current recommended methods for disk susceptibility testing and criteria for quality control testing.

A standard for global application developed through the Clinical and Laboratory Standards Institute consensus process.

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Clinical and Laboratory Standards Institute
P: +1.610.688.0100
F: +1.610.688.0700
www.clsi.org
standard@clsi.org

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Performance Standards for Antimicrobial Disk Susceptibility Tests

Barbara L. Zimmer, PhD
Darcie E. Carpenter, PhD, CIC, CEM
German Esparza, MSc
Kevin Alby, PhD, D(ABMM)
Amelia Bhatnagar, MPH
Andrea L. Ferrell, MLS(ASCP)CM
Laurie Flemming, SM, MT(ASCP)
Michael D. Huband, BS
Antonieta Jiménez-Pearson, MHA, PhD
Susan M. Kircher, MS, MT(ASCP)
Susan Weir, MPH, PhD
Rebecca Yee, PhD, D(ABMM)

Abstract

Clinical and Laboratory Standards Institute M02—*Performance Standards for Antimicrobial Disk Susceptibility Tests* includes a series of procedures to standardize the way disk diffusion tests are performed. The performance, applications, and limitations of the current CLSI-recommended methods are also described.

Antimicrobial susceptibility testing is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy, if its susceptibility cannot be reliably predicted from knowledge of the organism's identity. Susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents.

Various laboratory methods can be used to measure the *in vitro* susceptibility of bacteria to antimicrobial agents. In many medical microbiology laboratories, a disk diffusion method is used routinely for testing common, rapidly growing, and certain fastidious bacterial pathogens.

The supplemental information (CLSI M100[1] tables) used with this standard represents the most current information for drug selection, interpretation, and QC using the procedures standardized in CLSI M02.

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P: +1.610.688.0100 **F:** +1.610.688.0700 **E:** customerservice@clsi.org **W:** www.clsi.org

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Committee Membership

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Subcommittee on Antimicrobial Susceptibility Testing

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Working Group on AST Disk Susceptibility and Dilution AST for Aerobes (CLSI M02 and CLSI M07)

Barbara L. Zimmer, PhD
Chairholder
Beckman Coulter, Inc.
USA

Darcie E. Carpenter, PhD, CIC, CEM
Vice-Chairholder
International Health Management Associates, Inc.
USA

German Esparza, MSc
Vice-Chairholder
Proasecal SAS Colombia
Colombia

Kevin Alby, PhD, D(ABMM)
UNC Hospital
USA

Amelia Bhatnagar, MPH
Centers for Disease Control and Prevention
USA

Andrea L. Ferrell, MLS(ASCP)^{CM}
BD
USA

Laurie Flemming, SM, MT(ASCP)
National Institutes of Health, Department of Laboratory Medicine
USA

Michael D. Huband, BS
JMI Laboratories
USA

Antonieta Jiménez-Pearson, MHA, PhD
INCIENSA
Costa Rica

Susan M. Kircher, MS, MT(ASCP)
BD Diagnostic Systems
USA

Susan Weir, MPH, PhD
Global Microbiome Conservancy at Openbiome
USA

Rebecca Yee, PhD, D(ABMM)
The George Washington University
USA

Expert Panel on Microbiology

Expert panel volunteers support the development of CLSI documents by providing technical expertise in specialty areas. Expert panel members are listed by area of expertise on the CLSI website: https://clsi.org/standards-development/expert-panels/

Staff

Clinical and Laboratory Standards Institute
USA

Emily J. Gomez, MS, MLS(ASCP)^{CM}MB^{CM}
Program Manager

Laura Martin
Editorial Manager

Catherine E.M. Jenkins, ELS
Editor

Kristy L. Leirer, MS
Editor

Lisa M.W. Walker, MS, ELS
Editor

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Foreword

The most current edition of CLSI M100,[1] an annually published volume of tables, is made available with this standard to ensure users are aware of the latest recommendations related to the methods described in CLSI M02, M07,[2] and M11.[3]

Many other editorial and procedural changes in this edition of CLSI M02 resulted from Subcommittee on Antimicrobial Susceptibility Testing meetings held since 2018. Specific changes to the tables are summarized at the beginning of CLSI M100.[1] The most important changes in CLSI M02 are summarized below.

Overview of Changes

This standard replaces CLSI M02-Ed13, published in 2018. Several changes were made to this edition, including:

- **General:**
  - Revised information for testing and reporting to clarify relevant stakeholders
  - Revised nomenclature from "coagulase-negative staphylococci" (CoNS) to "staphylococci other than *Staphylococcus aureus*" (SOSA)
  - Revised nomenclature from "groups" to "tiers"
- **Background:**
  - Added information on US Food and Drug Administration disk clearance
- **Subchapter 1.4.1, Definitions:**
  - Added definitions for antimicrobial susceptibility testing, disk diffusion, EDTA-modified carbapenem inactivation method, heteroresistance, and reference method
  - Revised definitions for breakpoint, D-zone test, supplemental (not routine) test, and susceptible-dose dependent
- **Subchapter 1.4.2, Abbreviations and Acronyms:**
  - Added abbreviations for Mueller-Hinton fastidious and staphylococci other than *S. aureus*
  - Deleted abbreviation for coagulase-negative staphylococci
- **Subchapter 2.1, Selecting Antimicrobial Agents for Routine Testing and Reporting:**
  - Moved former Subchapter 2.4, Selection Guidelines, to new Subchapter 2.1, Selecting Antimicrobial Agents for Routine Testing and Reporting, with clarified and updated information on selecting agents for routine reporting
- **Subchapter 2.2, Equivalent Agents:**
  - Replaced former Subchapter 2.2, Routine Reports, with Subchapter 2.2, Equivalent Agents, with information on antimicrobial agents listed in CLSI M100[1] Tables 1
- **Subchapter 2.3, Suggested Guidelines for Routine and Selective Testing and Reporting:**
  - Revised suggested guidelines for routine and selective testing and reporting
  - Added new Table 2, Antimicrobial Agent Test and Report Tiers and Additional Considerations for Agents Listed in CLSI M100[1] Tables 1
  - Added new Table 3, Antimicrobial Agent Test and Report Designations and Additional Considerations for Agents Not Listed in CLSI M100[1] Tables 1
  - Deleted Subchapter 2.3.1, β-Lactams
  - Deleted Subchapter 2.3.2, Non-β-Lactams

- **Subchapter 2.4, Antimicrobial Classes:**
  - Added new Table 4, Antimicrobial Agent Class Characteristics
- **Subchapter 3.1, Disk Diffusion Test Reagents:**
  - Added information on cefiderocol disk diffusion testing
- **Subchapter 3.4.2, Colony Suspension Method for Inoculum Preparation:**
  - Revised colony suspension method for inoculum preparation
- **Subchapter 3.4.4, Performing Colony Counts:**
  - Added new subchapter with instructions for performing colony counts in step-action table
- **Subchapter 3.7, Reading Plates and Interpreting Results:**
  - Revised process for testing *Staphylococcus* spp. with oxacillin disk
- **Subchapter 3.8, Special Considerations for Fastidious Organisms:**
  - Added information on Mueller-Hinton fastidious agar for *Haemophilus influenzae* and *Streptococcus pneumoniae* disk diffusion testing
- **Subchapter 3.9.1, Special Considerations for Detecting Resistance in Staphylococci:**
  - Added information on *mecC* isolates
  - Renumbered former Table 2, Methods for Detecting Oxacillin Resistance in Staphylococci, as Table 6, Methods or Targets for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp., and revised incubation times and methods
  - Renumbered former Table 3, Procedural Recommendations for Detecting Oxacillin Resistance in Staphylococci, as Table 7, Procedural Recommendations for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.
  - Revised information on vancomycin-resistant *S. aureus,* vancomycin-intermediate *S. aureus,* and the vancomycin agar screen for *S. aureus*
  - Deleted former Subchapter 3.9.1.5, Linezolid Resistance, and instructions for reading linezolid disk diffusion test with transmitted light
- **Subchapter 3.9.2, Special Considerations for Detecting Resistance in Enterococci:**
  - Added information on detecting enterococci resistance with vancomycin agar screen test
- **Subchapter 3.9.4, Special Considerations for Detecting Resistance in Gram-Negative Bacilli:**
  - Added additional examples of β-lactamases in gram-negative bacteria to Table 8, Enzyme Classifications for β-Lactamases
  - Added additional examples of β-lactamases with carbapenemase activity to Table 9, β-Lactamases With Carbapenemase Activity
  - Revised information on AmpC enzymes and carbapenemases
- **Subchapter 3.11.3, Development of Resistance and Testing Repeat Isolates:**
  - Revised information on development of resistance
- **Subchapter 4.3, Selecting Strains for Quality Control:**
  - Revised information on selecting QC strains

- **Appendix A, Preparation of Media and Reagents:**
  - Added instructions for checking pH of GC agar to step-action table in section A1.3, GC Agar + 1% Defined Growth Supplement
  - Added instructions for checking pH of *Haemophilus* test medium agar to step-action table in section A1.4, *Haemophilus* Test Medium Agar
  - Added instructions for preparing Mueller-Hinton fastidious agar for testing *H. influenzae* or *S. pneumoniae* to new section A1.5, Mueller-Hinton Fastidious Agar
- **Appendix B, Conditions for Disk Diffusion Antimicrobial Susceptibility Tests:**
  - Added Mueller-Hinton fastidious agar for testing *H. influenzae* and *S. pneumoniae* to Table B2, Conditions for Disk Diffusion Antimicrobial Susceptibility Tests for Fastidious Organisms
- **Appendix E, Test for Performing Disk Diffusion Directly From Positive Blood Culture Broth:**
  - Added a procedure for performing disk diffusion directly from positive blood culture broth

Summary of CLSI Processes for Establishing Breakpoints and Quality Control Ranges

The Clinical and Laboratory Standards Institute (CLSI) is an international, voluntary, not-for-profit, interdisciplinary, standards-developing, and educational organization accredited by the American National Standards Institute that develops and promotes the use of consensus-developed standards and guidelines within the health care community. These consensus standards and guidelines are developed in an open and consensus-seeking forum to cover critical areas of diagnostic testing and patient health care. CLSI is open to anyone or any organization that has an interest in diagnostic testing and patient care. Information about CLSI can be found at www.clsi.org.

The CLSI Subcommittee on Antimicrobial Susceptibility Testing reviews data from a variety of sources and studies (eg, *in vitro,* pharmacokinetics/pharmacodynamics, and clinical studies) to establish antimicrobial susceptibility test methods, breakpoints, and QC parameters. The details of the data necessary to establish breakpoints, QC parameters, and how the data are presented for evaluation are described in CLSI M23.[4]

Over time, a microorganism's susceptibility to an antimicrobial agent may decrease, resulting in a lack of clinical efficacy and/or safety. In addition, microbiological methods and QC parameters may be refined to ensure more accurate and better performance of susceptibility test methods. Because of these types of changes, CLSI continually monitors and updates information in its documents. Although CLSI standards and guidelines are developed using the most current information available at the time, the field of science and medicine is always changing; therefore, standards and guidelines should be used in conjunction with clinical judgment, current knowledge, and clinically relevant laboratory test results to guide patient treatment.

Additional information, updates, and changes in this standard are found in the meeting summary minutes of the CLSI Subcommittee on Antimicrobial Susceptibility Testing at https://clsi.org/meetings/ast-file-resources/.

CLSI Subcommittee on Antimicrobial Susceptibility Testing Mission Statement

The CLSI Subcommittee on Antimicrobial Susceptibility Testing is composed of representatives from the professions, government, and industry, including microbiology laboratories, government agencies, health care providers and educators, and pharmaceutical and diagnostic microbiology industries. Using the CLSI voluntary consensus process, the subcommittee develops standards that promote accurate antimicrobial susceptibility testing and appropriate reporting. The mission of the CLSI Subcommittee on Antimicrobial Susceptibility Testing is to:

- Develop standard reference methods for antimicrobial susceptibility tests.
- Provide quality control parameters for standard test methods.
- Establish breakpoints and interpretive categories for the results of standard antimicrobial susceptibility tests and provide epidemiological cutoff values when breakpoints are not available.
- Provide suggestions for testing and reporting strategies that are clinically relevant and cost-effective.
- Continually refine standards and optimize detection of emerging resistance mechanisms through development of new or revised methods, breakpoints, and QC parameters.
- Educate users through multimedia communication of standards and guidelines.
- Foster a dialogue with users of these methods and those who apply them.

The ultimate purpose of the subcommittee's mission is to provide useful information to enable laboratories to assist the clinician in the selection of appropriate antimicrobial therapy for patient care. The standards and guidelines are meant to be comprehensive and to include all antimicrobial agents for which the data meet established CLSI guidelines. The values that guide this mission are quality, accuracy, fairness, timeliness, teamwork, consensus, and trust.

NOTE: The content of this standard is supported by the CLSI consensus process and does not necessarily reflect the views of any single individual or organization.

KEY WORDS

antibiotic

antimicrobial agents

disk diffusion

susceptibility testing

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Chapter 1

Introduction

Performance Standards for Antimicrobial Disk Susceptibility Tests

1 Introduction

1.1 Scope

This standard describes the reference disk diffusion method used to determine the *in vitro* antimicrobial susceptibility of bacteria that grow aerobically and includes:

- Agar plate preparation
- Testing conditions, including inoculum preparation and standardization, incubation time, and incubation temperature
- Results interpretation
- QC procedures
- Disk diffusion method limitations

To assist the medical laboratory, recommendations are provided for selecting antimicrobial agents for routine testing and reporting.

Standards for testing the *in vitro* antimicrobial susceptibility of bacteria that grow aerobically using dilution methods are found in CLSI M07.[2] Standards for testing the *in vitro* antimicrobial susceptibility of bacteria that grow anaerobically are found in CLSI M11.[3] Guidelines for standardized antimicrobial susceptibility testing (AST) of infrequently isolated or fastidious bacteria that are not included in CLSI M02, M07,[2] or M11[3] are available in CLSI M45.[5] The AST methods provided in this standard can be used in laboratories around the world, including but not limited to:

- Medical laboratories
- Public health laboratories
- Research laboratories
- Food laboratories
- Environmental laboratories

1.2 Background

Various laboratory methods can be used to measure the *in vitro* susceptibility of bacteria to antimicrobial agents. In many medical microbiology laboratories, a disk diffusion method is used routinely for testing common, rapidly growing, and certain fastidious bacterial pathogens. This standard describes the performance, applications, and limitations of the standardized disk diffusion test method. Recommendations[6] and governmental regulations[7,8] proposed by the US Food and Drug Administration (FDA) have been reviewed, and appropriate information is incorporated into this standard. Other AST methods, particularly disk diffusion reading devices, exist that provide results essentially equivalent to the CLSI methods described herein. The FDA is responsible for the clearance of antimicrobial agent disks and for the approval of commercial devices used in the United States, including specific devices for disk testing such as zone readers. CLSI does not approve or endorse commercial products or devices.

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Disk diffusion tests based solely on the presence or absence of a zone of inhibition without regard to the zone's size are not acceptable for determining antimicrobial susceptibility. Reliable results can be obtained only with disk diffusion tests that use standardized methodology and zone diameter measurements correlated with minimal inhibitory concentrations (MICs) with reference strains known to be susceptible, intermediate, or resistant to various antimicrobial agents.

The methods described herein must be followed explicitly to obtain accurate and reproducible results. This standardized method currently recommended by the CLSI Subcommittee on Antimicrobial Susceptibility Testing is based on the original and most thoroughly described disk diffusion method,[9] for which breakpoints and interpretive categories have been developed and supported by *in vitro* and clinical data.

This standard, along with CLSI M100,[1] describes methods, QC, breakpoints, and interpretive categories currently recommended for disk diffusion susceptibility tests. To develop these criteria for most antimicrobial agents, zone diameters are compared with MICs for many isolates, including those with known resistance mechanisms relevant to the particular drug class. Second, the MICs and correlated zone sizes are analyzed in relation to the pharmacokinetics of the drug from normal dosage regimens. Finally, when feasible, *in vitro* breakpoints are analyzed in relation to studies of clinical efficacy and microbiological eradication in the treatment of specific pathogens, as outlined in CLSI M23.[4]

When new limitations are recognized or improvements in these criteria are developed, changes will be incorporated into future editions of this standard and CLSI M100.[1]

1.3 Standard Precautions

Because it is often impossible to know what isolates or specimens might be infectious, all patient and laboratory specimens are treated as infectious and handled according to "standard precautions." Standard precautions are guidelines that combine the major features of "universal precautions and body substance isolation" practices. Standard precautions cover the transmission of all known infectious agents and thus are more comprehensive than universal precautions, which are intended to apply only to transmission of bloodborne pathogens. Published guidelines are available that discuss the daily operations of diagnostic medicine in humans and animals while encouraging a culture of safety in the laboratory.[10] For specific precautions for preventing the laboratory transmission of all known infectious agents from laboratory instruments and materials and for recommendations for the management of exposure to all known infectious diseases, refer to CLSI M29.[11]

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1.4 Terminology

CLSI, as a global leader in standardization, is firmly committed to achieving global harmonization whenever possible. Harmonization is a process of recognizing, understanding, and explaining differences while taking steps to achieve worldwide uniformity. CLSI recognizes that medical conventions in the global metrological community have evolved differently in different countries and regions and that legally required use of terms, regional usage, and different consensus timelines are all important considerations in the harmonization process. CLSI recognizes its important role in these efforts, and its consensus process focuses on harmonization of terms to facilitate the global application of standards and guidelines. Table 1 is provided to clarify the intended interpretations of common terms.

Table 1. Common Terms or Phrases With Intended Interpretations

Term or Phrase	Intended Interpretation
"Needs to" or "must"	Explains an action directly related to fulfilling a regulatory and/or accreditation requirement or is indicative of a necessary step to ensure patient safety or proper fulfillment of a procedure
"Require"	Represents a statement that directly reflects a regulatory, accreditation, performance, product, or organizational requirement or a requirement or specification identified in an approved documentary standard
"Should"	Describes a recommendation provided in laboratory literature, a statement of good laboratory practice, or a suggestion for how to meet a requirement

1.4.1 Definitions

For purposes of CLSI M02, the terms and definitions listed below apply. Consult CLSI's Harmonized Terminology Database at https://htd.clsi.org for related terms and definitions.

antimicrobial susceptibility testing (AST) – procedures used to determine the susceptibility of bacteria to a specific antimicrobial agent(s); **NOTE:** There are two major testing types, disk diffusion and dilution (broth and agar).

breakpoint – minimal inhibitory concentration (MIC) or zone diameter value used to categorize an organism as susceptible, susceptible-dose dependent, intermediate, resistant, or nonsusceptible; **NOTE 1:** MIC or zone diameter values generated by a susceptibility test can be interpreted based on established breakpoints; **NOTE 2:** Because breakpoints are largely based on pharmacologically and clinically rich datasets using *in vitro* and *in vivo* data, they are considered robust predictors of likely clinical outcome; **NOTE 3:** Also known as "clinical breakpoint"; **NOTE 4**: See **interpretive category.**

Carba NP test – colorimetric microtube assay for detecting carbapenemase production.[12,13]

disk diffusion – an *in vitro* antimicrobial susceptibility test conducted using disks impregnated with a specified single concentration of an antimicrobial agent applied to the surface of an agar medium that has been inoculated with the test organism; **NOTE:** The diameter of the zone of growth inhibition that results from the diffusion of an antimicrobial agent from the disks is measured with calipers or a ruler, recorded in millimeters, and interpreted according to CLSI standards.

D-zone test – a disk diffusion test using 15-µg erythromycin and 2-µg clindamycin disks spaced 15 to 26 mm apart for *Staphylococcus* spp. and 12 mm apart for *Streptococcus pneumoniae* and β-hemolytic *Streptococcus* spp. to detect the presence of inducible clindamycin resistance.

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EDTA-modified carbapenem inactivation method (eCIM) – assay used to differentiate metallo-β-lactamases from serine carbapenemases in Enterobacterales isolates that are positive for the modified carbapenem inactivation method (mCIM).

epidemiological cutoff value (ECV) – the minimal inhibitory concentration (MIC) or zone diameter value that separates microbial populations into those with and without phenotypically detectable resistance (non-wild-type or wild-type, respectively). The ECV defines the highest MIC or smallest zone diameter for the wild-type population of isolates.

EXAMPLE:

Category	Example of Categories for an ECV of 4 µg/mL and 19 mm	
	MIC, µg/mL	Zone Diameter, mm
Wild-type	≤ 4	≥ 20
Non-wild-type	≥ 8	≤ 19

- **wild-type** – an interpretive category defined by an ECV that describes the microbial population with no phenotypically detectable mechanisms of resistance or reduced susceptibility for the antimicrobial agent being evaluated.
- **non-wild-type** – an interpretive category defined by an ECV that describes the microbial population with phenotypically detectable mechanisms of resistance and reduced susceptibility for the antimicrobial agent being evaluated.

heteroresistance – presence of preexisting subpopulations of microbial cells exhibiting higher levels of resistance toward an antimicrobial agent while the majority of the cell population within the same culture is phenotypically susceptible to the same antimicrobial agent.

interpretive category – category derived from microbiological characteristics, pharmacokinetic/pharmacodynamic parameters, and clinical outcome data, when available; **NOTE 1:** Minimal inhibitory concentration (MIC) or zone diameter values generated by a susceptibility test can be interpreted based on established breakpoints; **NOTE 2:** See **breakpoint.**

EXAMPLE:

Interpretive Category	Breakpoints	
	MIC, µg/mL	Zone Diameter, mm
Susceptible	≤ 4	≥ 20
Susceptible-dose dependent	8–16	15–19
Intermediate	8–16	15–19
Resistant	≥ 32	≤ 14
Nonsusceptible	> 1	< 17

MIC or zone diameter value breakpoints and interpretive categories are established per CLSI M23[4] for categories of susceptible, intermediate, and resistant (and susceptible-dose dependent and nonsusceptible, when appropriate). CLSI susceptible (S) or susceptible-dose dependent (SDD) breakpoints added or revised since 2010 have been based on specific dosage regimen(s); these dosage regimens are listed in CLSI M100[1] Table 2, Dosages. Antimicrobial Agent Dosage Regimens Used to Establish Susceptible or Susceptible-Dose Dependent Breakpoints (referred to as "CLSI M100[1] Table 2 Dosages").

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- **susceptible (S)** – a category defined by a breakpoint that implies that isolates with an MIC at or below or a zone diameter at or above the susceptible breakpoint are inhibited by the usually achievable concentrations of antimicrobial agent when the dosage recommended to treat the site of infection is used, resulting in likely clinical efficacy.
- **susceptible-dose dependent (SDD)** – a category defined by a breakpoint that implies that susceptibility of an isolate depends on the dosage regimen that is used in the patient. To achieve levels that are likely to be clinically effective against isolates for which the susceptibility testing results (either MICs or zone diameters) are in the SDD category, it is necessary to use a dosage regimen (ie, higher doses, more frequent doses, or both or extended infusion) that results in higher drug exposure than that achieved with the dose that was used to establish the susceptible breakpoint. Consideration should be given to the maximum, literature-supported dosage regimen, because higher exposure gives the highest probability of adequate coverage of an SDD isolate. CLSI M100[1] Table 2 Dosages list the doses used when establishing SDD categories. The drug label should be consulted for recommended doses and adjustment for organ function; **NOTE:** The SDD category may be assigned when doses well above those used to calculate the susceptible breakpoint are supported by the literature, widely used clinically, and/or approved and for which sufficient data to justify the designation exist and have been reviewed. This category also includes a buffer zone for inherent variability in test methods, which should prevent small, uncontrolled, technical factors from causing major discrepancies in interpretations, especially for drugs with narrow pharmacotoxicity margins. See CLSI M100[1] appendixes for additional information.
- **intermediate (I)** – a category defined by a breakpoint that includes isolates with MICs or zone diameters within the intermediate range that approach usually attainable blood and tissue levels and/or for which response rates may be lower than for susceptible isolates; **NOTE:** An I with a ^ in CLSI M100[1] Tables 2 indicates agents that have the potential to concentrate in the urine. The I^ is for informational use only. The decision to report I^ is best made by each laboratory based on institution-specific guidelines and in consultation with appropriate medical personnel. The I category also includes a buffer zone for inherent variability in test methods, which should prevent small, uncontrolled, technical factors from causing major discrepancies in interpretations, especially for drugs with narrow pharmacotoxicity margins.
- **resistant (R)** – a category defined by a breakpoint that implies that isolates with an MIC at or above or a zone diameter at or below the resistant breakpoint are not inhibited by the usually achievable concentrations of the agent with normal dosage schedules and/or that demonstrate MICs or zone diameters that fall in the range in which specific microbial resistance mechanisms are likely, and clinical efficacy of the agent against the isolate has not been reliably shown in treatment studies.
- **nonsusceptible (NS)** – a category used for isolates for which only a susceptible breakpoint is designated because of the absence or rare occurrence of resistant strains. Isolates for which the antimicrobial agent MICs are above or the zone diameters are below the value indicated for the susceptible breakpoint should be reported as nonsusceptible; **NOTE 1:** An isolate that is interpreted as nonsusceptible does not necessarily mean that the isolate has a resistance mechanism. It is possible that isolates with MICs above the susceptible breakpoint that lack resistance mechanisms may be encountered within the wild-type distribution after the time the susceptible-only breakpoint was set; **NOTE 2:** The term "nonsusceptible" should not be used when the text is describing an organism/drug category with intermediate and resistant interpretive categories. Isolates that are in the categories of "intermediate" or "resistant" could be called "not susceptible" rather than "nonsusceptible."

minimal inhibitory concentration (MIC) – the lowest concentration of an antimicrobial agent that prevents visible growth of a microorganism in an agar or broth dilution susceptibility test.

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modified carbapenem inactivation method (mCIM) – phenotypic assay for detecting carbapenemase production in which the ability of an isolate to degrade meropenem in a 10-µg disk is assessed.

quality assurance (QA) – **1)** part of quality management focused on providing confidence that quality requirements will be fulfilled[14]; **2)** a comprehensive set of policies, procedures, and practices used to monitor the laboratory's entire testing process and ensure that the testing site's results are reliable; **NOTE:** The practice that encompasses all procedures and activities directed toward ensuring that a specified quality of product or service is achieved and maintained. In the testing environment, this process includes monitoring all the raw materials, supplies, instruments, and procedures; sample collection, transport, storage, and processing; recordkeeping; calibrating and maintaining equipment; quality control; proficiency testing; training of personnel; and all else involved in the production of the data reported.

quality control (QC) – part of quality management focused on fulfilling quality requirements[14]; **NOTE:** A system for ensuring maintenance of proper standards by periodic inspection of the results and the operational techniques that are used to ensure accuracy and reproducibility.

reference method – a thoroughly investigated method in which exact and clear descriptions of the necessary conditions and procedures are given for the accurate determination of one or more property values, and in which the documented trueness and precision of the method are commensurate with the method's use for assessing the trueness of other methods for measuring the same property values or for assigning reference method values to reference materials.

routine test – disk diffusion or broth or agar dilution minimal inhibitory concentration tests for routine clinical testing.

saline – a solution of 0.85% to 0.9% NaCl (w/v).

screening test – a test that provides presumptive results; **NOTE:** Additional testing is typically needed only for a specific result (eg, only when the screening test result is positive).

supplemental (not routine) test – test that detects susceptibility or resistance to an antimicrobial agent or drug class by a method other than routine disk diffusion or broth or agar dilution and does not need additional tests to confirm susceptibility or resistance; **NOTE 1:** Some supplemental tests identify a specific resistance mechanism; **NOTE 2:** A supplemental test may be required or optional for reporting specific clinical results.

surrogate agent test – test performed with an agent that replaces a test performed with the antimicrobial agent of interest and is used when the agent of interest cannot be tested because of availability or performance issues (eg, surrogate agent performs better than the agent of interest).

test method – the method (ie, either the routine laboratory method or automated method) that is compared with the reference method.

test system – system that includes instructions and all the instrumentation, equipment, reagents, and/or supplies needed to perform an assay or examination and generate test results.

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1.4.2 Abbreviations and Acronyms

AST	antimicrobial susceptibility testing
ATCC®[a]	American Type Culture Collection
BHI	brain heart infusion
CFU	colony-forming unit(s)
CSF	cerebrospinal fluid
eCIM	EDTA-modified carbapenem inactivation method
EDTA	ethylenediaminetetraacetic acid
ESBL	extended-spectrum β-lactamase
FDA	US Food and Drug Administration
HTM	*Haemophilus* test medium
hVISA	heteroresistant vancomycin-intermediate *Staphylococcus aureus*
mCIM	modified carbapenem inactivation method
MHA	Mueller-Hinton agar
MH-F	Mueller-Hinton fastidious
MIC	minimal inhibitory concentration
MRS	methicillin (oxacillin)-resistant staphylococci
MRSA	methicillin (oxacillin)-resistant *Staphylococcus aureus*
PBP	penicillin-binding protein
PBP2a	penicillin-binding protein 2a
PBP2c	penicillin-binding protein 2c
pH	negative logarithm of hydrogen ion concentration
QA	quality assurance
QC	quality control
RNA	ribonucleic acid
SDD	susceptible-dose dependent
SOSA	staphylococci other than *Staphylococcus aureus*
VRE	vancomycin-resistant enterococci

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Chapter 2

Indications for Performing Antimicrobial Susceptibility Tests

2 Indications for Performing Antimicrobial Susceptibility Tests

AST is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy if its susceptibility cannot be reliably predicted from knowledge of the organism's identity (eg, intrinsic resistance). Antimicrobial susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents. Resistance mechanisms may include:

- Production of drug-inactivating enzymes
- Alteration of drug targets
- Altered drug uptake or efflux

Some organisms have predictable susceptibility to antimicrobial agents, and empiric therapy for these organisms is widely accepted. Antimicrobial susceptibility tests are seldom necessary when the infection is caused by a microorganism recognized as susceptible to a highly effective drug (eg, the continued susceptibility of *Streptococcus pyogenes* to penicillin). Antimicrobial susceptibility tests are also important in epidemiological studies of resistance, in studies of new antimicrobial agents, and in antimicrobial resistance surveillance.

When the nature of the infection is not clear and the specimen contains mixed growth or normal flora (in which the organisms probably bear little relationship to the infectious process being treated), antimicrobial susceptibility tests are often unnecessary, and the results may be misleading.

Isolated colonies of each organism type that may be pathogenic should be selected from primary agar plates and tested individually for susceptibility. Identification procedures are often performed at the same time. Mixtures of different microorganism types should not be tested on the same susceptibility test.

2.1 Selecting Antimicrobial Agents for Routine Testing and Reporting

To make routine AST relevant and practical, the antimicrobial agents tested should be limited. CLSI M100[1] Tables 1 list antimicrobial agents that fulfill the basic needs for routine use in most medical laboratories. The tables are based on specific organisms or organism groups, with the various antimicrobial agents indicated in tiers for testing to assist laboratories in selecting their routine testing batteries.

Selecting the most appropriate antimicrobial agents to test and report is a decision best made by each laboratory in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, as appropriate. Care should be taken to include the specific antimicrobial agents in the formulary as well as those that physicians prescribe daily.

The suggestions for each organism group in CLSI M100[1] Tables 1 include agents of proven efficacy that show acceptable *in vitro* test performance. Considerations in the assignment of agents to specific tiers include:

- Clinical efficacy
- Prevalence of resistance
- Minimizing emergence of resistance
- FDA clinical indications for use
- Current consensus recommendations for first-choice and alternative drugs
- Cost

Tests on selected agents may be useful for infection-prevention purposes.

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2.2 Equivalent Agents

Antimicrobial agents listed together in a single box in CLSI M100[1] Tables 1 are agents for which interpretive categories (susceptible, intermediate, susceptible-dose dependent [SDD], nonsusceptible, or resistant) and clinical efficacy are similar. A laboratory will often test only one agent from a box routinely, typically the agent that is on its formulary. In some cases, a laboratory may not test any agents from a box, depending on institutional needs.

In some boxes, agents will be listed with an "or" between them. The "or" identifies agents for which cross-resistance and cross-susceptibility are nearly complete. Results from one agent connected by an "or" can be used to predict results for the other agent (ie, equivalent agents). For example, Enterobacterales susceptible to cefotaxime can be considered susceptible to ceftriaxone. The results obtained from testing cefotaxime could be reported along with a comment that the isolate is also susceptible to ceftriaxone. For drugs connected with an "or," combined major and very major errors are fewer than 3%, and minor errors are fewer than 10%, based on a large population of bacteria tested (see CLSI M23[4] for description of error types). In addition, to qualify for an "or," at least 100 strains with resistance to the agents in question must be tested, and a result of "resistant" must be obtained with all agents for at least 95% of the strains. "Or" is also used for comparable agents when tested against organisms for which "susceptible-only" breakpoints are provided (eg, cefotaxime or ceftriaxone with *Haemophilus influenzae*). When no "or" connects agents within a box, testing of one agent cannot be used to predict results for another, owing either to discrepancies or insufficient data (see CLSI M100[1] Instructions for Use of Tables).

2.3 Suggested Guidelines for Routine and Selective Testing and Reporting

The recommended antimicrobial agents are considered appropriate for testing and reporting following the recommendations in CLSI M100[1] Tables 1. The recommendations in CLSI M100[1] Tables 1 for each organism group include antimicrobial agents of proven efficacy that show acceptable *in vitro* test performance. Test and report Tiers 1, 2, 3, 4 and "urine only" are noted in CLSI M100[1] Tables 1. CLSI M100[1] Tables 2 contain two additional test and report tier designations, "other" and "investigational."

Antimicrobial agents other than those appropriate for use in therapy may also be tested to provide taxonomic data (eg, colistin for identification purposes) and epidemiological information (eg, testing streptomycin vs *Salmonella* by public health laboratories for surveillance purposes), but they should not be included on patient reports. However, such results should be available to the infection control practitioner and/or hospital epidemiologist.

Each laboratory should decide which antimicrobial agents in CLSI M100[1] Tables 1 to routinely report (Tier 1) and which might be reported only selectively (Tiers 2 to 4). Each laboratory should consider developing these selective and/or cascade reporting rules in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, as appropriate. Selective and cascade reporting is done to encourage appropriate antimicrobial agent use. The positioning of drugs in tiers in CLSI M100[1] Tables 1 can be used to guide development of routine, selective, and/or cascade reporting rules. In addition, each laboratory should develop a protocol to test additional antimicrobial agents on isolates that are confirmed as resistant to all antimicrobial agents on its routine test panel. This protocol should include options for testing additional agents in-house or sending the isolate to a referral laboratory.

Selective reporting involves reporting results for specific antimicrobial agents based on defined criteria unrelated to results obtained from AST (eg, organism identification, body site, clinical setting, or patient demographics) (see CLSI M100[1] for examples).

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Cascade reporting involves reporting results for specific agents based on the overall antimicrobial susceptibility profile of an isolate. Results for secondary or broader-spectrum agents (eg, Tier 2 or 3) are reported only if the isolate is resistant to primary or narrower-spectrum agents (eg, Tier 1) (see CLSI M100[1] for examples). A "resistant" result for a broader-spectrum agent (eg, Tier 2) should always be reported even if the organism tests "susceptible" to the narrower-spectrum agent (eg, Tier 1). Such unexpected resistant results should be confirmed (see CLSI M100[1] Tables 1 and CLSI M100[1] appendixes).

Cascade rules can be created for agents within the same tier or between tiers. Agents listed in the same row between tiers in CLSI M100[1] Tables 1 can be used as a guide for creating cascade reporting rules.

Table 2 describes the antimicrobial agent test and report tiers and additional considerations for agents listed in CLSI M100[1] Tables 1. Table 3 includes the test and report designations and additional considerations for agents not listed in CLSI M100[1] Tables 1.

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Table 2. Antimicrobial Agent Test and Report Tiers and Additional Considerations for Agents Listed in CLSI M100[1] Tables 1

Tier	Definition	Test	Report[a]	Additional Testing and Reporting Considerations
1	Antimicrobial agents that are appropriate for routine primary testing and reporting	Routine	Routine	
2	Antimicrobial agents that are appropriate for routine primary testing but may be reported following cascade reporting rules established at each institution	Routine	Cascade	• Report following cascade reporting rules because of resistance to agent(s) in Tier 1. • May be reported routinely based on institution-specific guidelines.
3	Antimicrobial agents that are appropriate for routine primary testing in institutions that serve patients at high risk for MDROs but should only be reported following cascade or selective reporting rules[b]	Routine or by request	Cascade	Test routinely based on institution-specific guidelines or by clinician request and report following cascade reporting rules because of resistance to agent(s) in Tiers 1 and 2.
4	Antimicrobial agents that may warrant testing and reporting by clinician request if antimicrobial agents in other tiers are not optimal because of various factors	By request	By request	• Test and report by clinician request because of: – Unavailability of preferred drug for clinical use – Patient underlying condition, including allergies – Unusual susceptibility profile of the organism, including resistance to agents in Tiers 1, 2, and 3 – Polymicrobial infection • May also warrant testing as an epidemiological aid
Urine only	Antimicrobial agents designated by a "(U)" in CLSI M100[1] Tables 2 should be reported only on organisms isolated from the urinary tract.	Routine	Report as appropriate	Agents in Tiers 1, 2, and 3 may also be reported on urine isolates, as appropriate, following the testing and reporting guidance for the respective tiers.

Abbreviations: MDRO, multidrug-resistant organism; UTI, urinary tract infection.
[a] Antimicrobial agents should be reported selectively, as appropriate (eg, because it is effective in treating uncomplicated UTIs only, nitrofurantoin would be reported only on isolates from urine).
[b] Identification of patients at high risk for MDROs will likely be communicated by infection preventionists. For examples of criteria used to identify patients at high risk for MDROs, see https://www.cdc.gov/hai/organisms/ESBL.html and https://www.cdc.gov/mrsa/community/index.html.

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Table 3. Antimicrobial Agent Test and Report Designations and Additional Considerations for Agents Not Listed in CLSI M100[1] Tables 1

Designation	Definition	Test	Report[a]	Additional Testing and Reporting Considerations
Other	Antimicrobial agents with established clinical breakpoints designated by * in CLSI M100[1] Tables 2 that are generally not candidates for testing and reporting in the United States	By request	By request	• Test and report only by clinician request and only following consultation with the antimicrobial stewardship team and other relevant institutional stakeholders to ensure appropriateness of the request. • Agents with an "other" designation may not reflect current consensus recommendations for first-choice and alternative drugs for the specific organism or organism group.
Inv.	Antimicrobial agents that are investigational for the organism group designated by "Inv." in CLSI M100[1] Tables 2 have not yet been approved by the FDA for use in the United States.	By request	By request	Test and report only by clinician request and only following consultation with the antimicrobial stewardship team and other relevant institutional stakeholders to ensure appropriateness of the request. These agents would likely be clinically available for compassionate use only.

Abbreviations: FDA, US Food and Drug Administration; inv., investigational.
[a] Antimicrobial agents should be reported selectively, as appropriate (eg, daptomycin should not be reported on isolates recovered from the respiratory tract because the agent interacts with pulmonary surfactant, resulting in inhibition of antibacterial activity).

2.4 Antimicrobial Classes

To minimize confusion, all antimicrobial agents should be reported using official nonproprietary (generic) names. The many currently available antimicrobial agents may be grouped together by drug classes, emphasizing their relatedness.

Table 4 provides brief descriptions of common antimicrobial agent classes, separated into β-lactams and non–β-lactams. For a complete list, see CLSI M100[1] glossaries.

Table 4. Antimicrobial Agent Class Characteristics

β-Lactams[a]
Description: • Share a common, central, 4-member β-lactam ring. • Majority bind to at least 2 PBPs. • Inhibit the transpeptidation step in bacterial cell wall synthesis. • Classification as a penicillin, β-lactam combination agent, cephem, monobactam, or penem determined by ring structures or substituent groups added to the β-lactam ring.

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Table 4. (Continued)

β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Penicillins	• Primarily active against non–β-lactamase–producing gram-positive aerobes, some fastidious gram-negative aerobes, and some anaerobic bacteria. • Penicillinase-stable penicillins are not hydrolyzed by penicillinase-producing staphylococci.	• Penicillinase-labile penicillins, which are divided into subclasses: penicillin, aminopenicillins, carboxypenicillins, amdinocillin, and ureidopenicillins • Penicillinase-stable penicillins
β-Lactam combination agents	• β-Lactam class antimicrobial agent plus second agent that functions as an inhibitor of some β-lactamases, potentiating the activity of the β-lactam antimicrobial agent • Broad-spectrum activity against gram-positive and gram-negative bacteria • Organisms that test susceptible to the β-lactam agent alone are also considered susceptible to the β-lactam combination agent. However, organisms that test susceptible to the β-lactam combination agent cannot be assumed to be susceptible to the β-lactam agent alone. Similarly, organisms that test SDD, intermediate, or resistant to the β-lactam agent alone may be susceptible to the β-lactam combination agent.	
Cephems	• Spectrum of activity depends on subclass (gram-positive, gram-negative, anaerobes). • Generally bind with the greatest affinity to PBP3 in enteric bacteria in addition to PBP1 and/or PBP1b. • Not all agents within a specific group or generation have the same spectrum of activity. • Because of differing activity, representatives of each group may be selected for routine testing. • Cephems are also separated based on oral or parenteral delivery.	• Often referred to as first-, second-, third-, or fourth-generation cephalosporins – Third- and fourth-generation cephalosporins are oxyimino-cephalosporins and sometimes referred to as extended-spectrum cephems. • Cephalosporins with anti-MRSA activity: cephamycins, oxycephem, and siderophore cephalosporins
Monobactams	• Monocyclic β-lactams • High affinity to PBP3 in gram-negative aerobic bacteria • Not hydrolyzed by metallo-β-lactamases	
Penems	• Differ slightly in structure from penicillins. • More stable than penicillins and cephems against β-lactamase hydrolysis. • Broad-spectrum activity against many gram-positive and gram-negative bacteria. • Hydrolyzed by class A, B, and D carbapenemases.	Carbapenems

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Table 4. (Continued)

Non–β-Lactams[a]		
Class	**Characteristics**	**Subclasses**
Aminoglycosides	• Interfere with protein synthesis by irreversibly binding to 16S ribosomal RNA within the 30S ribosomal subunit. • Variously affected by AMEs, resulting in some differences in the spectrum of activity. • Plazomicin is more stable against common AMEs; however, it is affected by 16 RNA methyl transferases. • Active only in aerobic conditions.	
Ansamycins	• Lipophilic macrocyclic antibiotic that easily diffuses through membranes • Aromatic moiety bridged by an aliphatic chain • Antibacterial action results from high-affinity binding to the DNA-dependent RNA polymerase of prokaryotes (RNAP) and inhibition of RNA synthesis. • Broad-spectrum activity against many gram-positive bacteria, gram-negative bacteria, and mycobacteria.	Rifamycins
DNA-gyrase and topoisomerase inhibitors	• Inhibit DNA-gyrase or topoisomerase activity. • Active against many gram-positive and gram-negative bacteria and some anaerobes and fastidious microorganisms. • Spectrum differences necessitate separate testing of the individual antimicrobial agents.	• Quinolones, fluoroquinolones, benzoquinolizine, spiropyrimidinetrione, and triazaacenaphthylenes • Spiropyrimidinetriones (eg, zoliflodacin) are active against *Neisseria gonorrhoeae*. Triazaacenaphthylenes (eg, gepotidacin) are active against pathogens causing uncomplicated UTI against *N. gonorrhoeae*. These drugs bind different targets and are less affected by common mechanisms of resistance.
Folate pathway antagonists	• Antimetabolic chemotherapeutic antimicrobial agents • Inhibit sequential steps in the folate pathway. • Active against some gram-positive and gram-negative bacteria. • Examples: – Trimethoprim: inhibits dihydrofolate reductase, which catalyzes the formation of tetrahydrofolate from dihydrofolate – Sulfamethoxazole: inhibits dihydropteroate synthase, which catalyzes the formation of dihydrofolate from *para*-aminobenzoic acid • Usually tested in combination: trimethoprim-sulfamethoxazole	Dihydrofolate reductase inhibitors and sulfonamides

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Table 4. (Continued)

Non–β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Glycopeptides	• Linear, cross-linked heptapeptides with a variable number of sugar moieties; may have lipid substituents. • Block subunit cross-linking during cell wall formation by binding to the D-alanyl-D-alanine termini in the stem peptide of nascent peptidoglycan. • Used for treating infections caused by aerobic gram-positive bacteria, especially β-lactam–resistant, gram-positive bacteria (eg, MRSA and some enterococci). • Used for treating gram-positive infections in penicillin-allergic patients. • Used as an oral treatment for *Clostridioides* (formerly *Clostridium*) *difficile* infection (ie, vancomycin).	Glycopeptides, lipoglycopeptides, and lipoglycodepsipeptides
Lipopeptides	• Target the cell membrane. • May be linear or cyclic. • Strongly influenced by the presence of divalent cations in the medium used in testing. • Polymyxins: – Active against gram-negative organisms only. – Bind to the outer cell membrane, leading to membrane-permeability changes and cell death. – Bind to and neutralize lipopolysaccharide. – Excess calcium cations inhibit activity. • Lipopeptides: – Active against gram-positive organisms only. – Form a calcium complex and insert into the cell membrane. – Disrupt cell membrane function with potassium efflux and membrane depolarization. – Presence of physiological calcium ion levels (50 mg/L) essential for optimal activity.	Polymyxins

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Table 4. (Continued)

Non–β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Macrocyclic lactone	Inhibit RNA synthesis by bacterial RNA polymerase at a site distinct from that of rifamycins, eg, fidaxomicin.	
Macrolides	• Inhibit bacterial protein synthesis by binding to the 23S RNA of the 50S ribosomal subunit. • Active against gram-positive and some gram-negative bacteria (eg, *Salmonella* and *Shigella* spp.). • Active against some fastidious bacteria (eg, *Haemophilus* spp.).	Ketolides and fluoroketolides
Nitroheterocyclics	• Nitrofurans: – Reduced intracellularly to a reactive metabolite that binds to macromolecules. – Used extensively to treat uncomplicated UTIs. • Nitroimidazoles: – Converted intracellularly in susceptible organisms to metabolites that bind to DNA and other cellular macromolecules. – Metronidazole is active only against strictly anaerobic bacteria and some protozoan parasites.	Nitrofuran, nitroimidazoles, and thiazolides
Oxazolidinones	• Inhibit protein synthesis • Binds to the 50S ribosome subunit, preventing peptide chain initiation • Active against gram-positive organisms and mycobacteria	
Phosphonates	• Inhibit bacterial cell wall by inactivating the enzyme UDP-N-acetylglucosamine-3-3-enolpryuvyltransferase (MurA). • Broad-spectrum activity against both gram-positive and gram-negative bacteria. • CLSI breakpoints for fosfomycin are established for *Escherichia coli* and *Enterococcus faecalis*. • Fosfomycin-trometamol is mainly used to treat uncomplicated UTIs.	Fosfomycin
Tetracyclines	• Inhibit protein synthesis by binding to the 30S ribosomal subunit and blocking binding of aminoacyl transfer RNA. • Active in certain gram-positive and gram-negative bacteria. – Glycylcyclines (tigecycline) and fluorocyclines (eravacycline) are also active against anaerobic bacteria. • Closely related group, and with few exceptions, only tetracycline might need to be tested routinely.	Fluorocycline, glycylcycline, and aminomethylcycline

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Table 4. (Continued)

Other Non–β-Lactam Drug Classes With One or Two Agents[a]		
Class	**Characteristics**	**Subclasses**
Protein synthesis inhibitors	Protein synthesis inhibition by several mechanisms like binding to the 50S ribosomal subunit and blocking the path by which nascent peptides exit the ribosome (eg, lincosamides) or binding to the 16S rRNA helix 34 of the 30S subunit of the bacterial ribosome, blocking the translocation step of protein synthesis (eg, spectinomycin)	• Aminocyclitols (ie, spectinomycin) • Lincosamides (ie, clindamycin) • Phenicols (ie, chloramphenicol) • Pleuromutilins (ie, lefamulin) • Pseudomonic acid (ie, mupirocin) • Quinolonyl oxazolidinone (ie, cadazolid) • Steroid (ie, fusidic acid) • Streptogramins (ie, quinupristin-dalfopristin)
Lysins	• Hydrolytic enzyme that cleaves the bacterial cell wall • Often bacterial species specific, eg, exebacase	
Peptide	Peptide antibiotics interfere with cell wall development by blocking the attachment of new cell wall subunits (eg, muramyl pentapeptides).	Magainin

Abbreviations: AME, aminoglycoside-inactivating enzyme; DNA, deoxyribonucleic acid; MRSA, methicillin (oxacillin)-resistant *Staphylococcus aureus*; PBP, penicillin-binding protein; RNA, ribonucleic acid; rRNA, ribosomal ribonucleic acid; SDD, susceptible-dose dependent; UTI, urinary tract infection.

[a] Refer to CLSI M100[1] glossaries.

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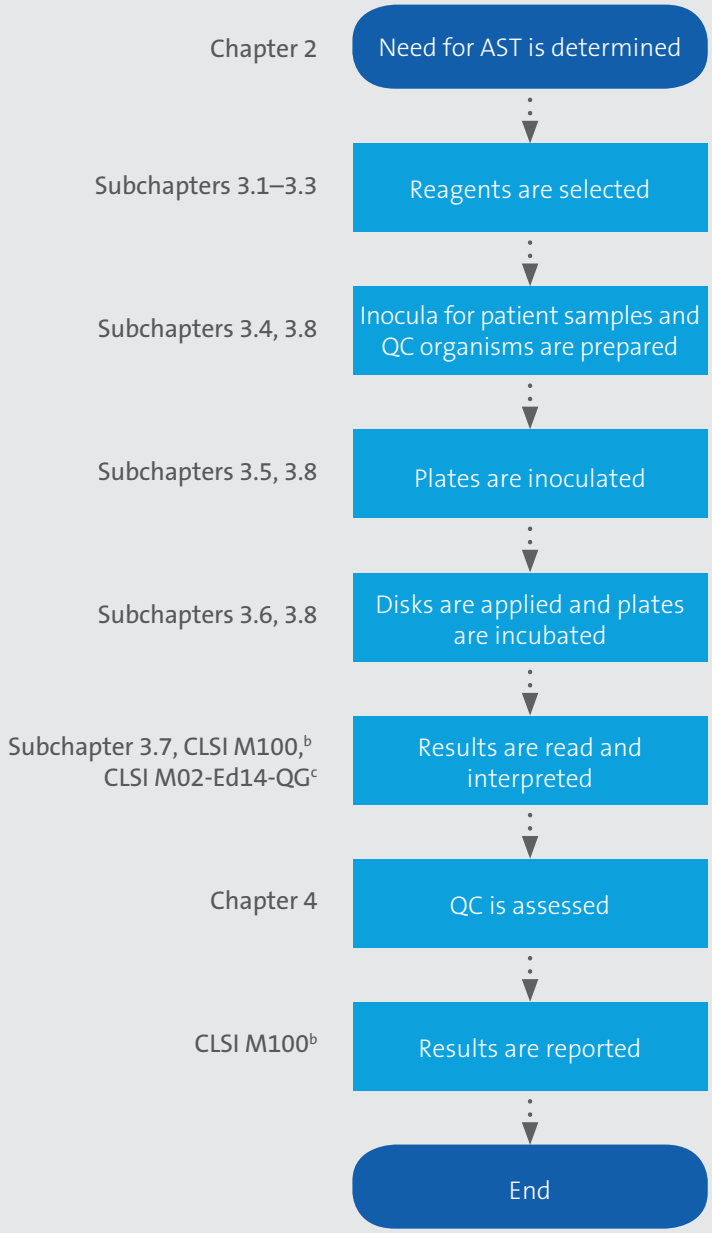
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Chapter 3

Disk Diffusion Antimicrobial Susceptibility Testing Process

3 Disk Diffusion Antimicrobial Susceptibility Testing Process

Figure 1 provides an overview of the disk diffusion AST process. Detailed information for each step is provided in each designated chapter or subchapter.



Abbreviations: AST, antimicrobial susceptibility testing; QC, quality control.
[a] Three basic symbols are used in this process flow chart: oval (signifies the beginning or end of a process), arrow (connects process activities), box (designates process activities).
[b] See CLSI M100[1] Tables 2.
[c] See CLSI *M02 Disk Diffusion Reading Guide*.[15]

Figure 1. Disk Diffusion AST Process[a]

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3.1 Disk Diffusion Test Reagents

3.1.1 Mueller-Hinton Agar

Mueller-Hinton agar (MHA) is considered the best medium for routine AST of nonfastidious bacteria because:

- It shows acceptable batch-to-batch reproducibility for AST.
- It is low in inhibitors that affect sulfonamide, trimethoprim, cefiderocol, and tetracycline susceptibility test results.
- It supports satisfactory growth of most nonfastidious pathogens.
- A substantial amount of data and experience has been collected about antimicrobial susceptibility tests performed with this medium.

Although MHA is generally reliable for AST, results obtained with some brands and batches may, on occasion, vary significantly. If a batch of medium does not support adequate growth of a test organism, zones will likely be erroneously large, and results with QC strains may exceed the acceptable QC limits. Commercially prepared plates may be used, or they may be prepared as described in Appendix A. Only MHA formulations that have been tested according to and that meet the acceptance limits as defined by the International Organization for Standardization[16] should be used.

3.1.1.1 pH

The agar medium should have a pH between 7.2 and 7.4 at room temperature. The method to assess pH is provided in Appendix A, section A1.1.

3.1.1.2 Effects of Thymidine or Thymine

MHA containing excessive amounts of thymidine or thymine can reverse the inhibitory effect of sulfonamides and trimethoprim, thus yielding smaller and less distinct zones, or no zone at all, which may result in false-resistance reports. The laboratory should use MHA that is as low in thymidine content as possible. If problems with sulfonamide and trimethoprim QC occur, it might be necessary to check the MHA. To evaluate an MHA lot, *E. faecalis* ATCC® 29212 or, alternatively, *E. faecalis* ATCC® 33186 may be tested with trimethoprim-sulfamethoxazole disks. Satisfactory media provide essentially clear, distinct zones of inhibition that are ≥ 20 mm. Unsatisfactory media produce no zone of inhibition, growth within the inhibition zone, or a zone of inhibition < 20 mm.

3.1.1.3 Effects of Variation in Divalent Cations

Variation in divalent cations, principally magnesium and calcium, affects results of aminoglycoside tests with *Pseudomonas aeruginosa* strains. Excess cation content reduces zone sizes, whereas low cation content may result in unacceptably large zones of inhibition. Variation in calcium levels also affects daptomycin test results. For daptomycin, insufficient calcium content reduces zone sizes, whereas high calcium content may increase zone sizes; therefore, disk diffusion testing is not reliable for testing daptomycin. Excess zinc ions may reduce carbapenem zone sizes. Iron concentrations affect cefiderocol broth microdilution testing and iron-depleted cation-adjusted Mueller-Hinton broth is needed. Iron depletion of MHA is not required for cefiderocol disk diffusion testing.[17] Performance tests with each MHA lot must conform to the QC limits listed in CLSI M100[1] Tables 4.

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3.1.2 Handling and Storing Mueller-Hinton Agar Plates

The steps for handling and storing MHA plates are listed in the following table.

Step	Action	Comment
1	Store prepared MHA plates at 2°C to 8°C in sealed packages until needed.	Storing plates in sealed packages prevents dehydration.
2	Remove the MHA plates from the refrigerator at least 15 minutes before use to let them warm to room temperature.	Letting plates warm to room temperature enables excess moisture to be more readily observed. Excess moisture could interfere with plate inoculation.
3	If agar surfaces contain excess moisture (eg, large condensation droplets), place MHA plates in an incubator (35°C ± 2°C) or a laminar flow hood at room temperature with lids ajar until excess surface moisture is removed by evaporation (usually 10–30 minutes).	• The plate surfaces should be moist, but no moisture droplets should be apparent on the medium's surface or on the Petri plate covers when the plates are inoculated. • Exercise caution to prevent overdrying of media.

Abbreviation: MHA, Mueller-Hinton agar.

3.2 Testing Strains That Fail to Grow Satisfactorily

Only aerobic or facultative bacteria that grow well on unsupplemented MHA should be tested on that medium. Certain fastidious species, such as *H. influenzae, Haemophilus parainfluenzae, Neisseria gonorrhoeae, Neisseria meningitidis,* and streptococci, do not grow sufficiently on unsupplemented MHA. These organisms need supplements or different media to grow, and they should be tested on the media listed below and described in Appendix A using methods described in this standard:

- *Haemophilus* test medium (HTM) and Mueller-Hinton fastidious (MH-F) agar
- GC agar base + 1% defined growth supplement
- MHA with 5% sheep blood

Details for these tests are provided in Subchapter 3.8 and Appendix B.

3.3 Antimicrobial Disks

3.3.1 Disk Source and Quality Specifications

The CLSI- and EUCAST-recommended procedure for determining the optimum concentration of a specific antimicrobial agent to incorporate into a disk when developing a disk diffusion method for that agent can be found in CLSI M23S.[18]

Disks should be purchased from a reliable commercial vendor. The disks should be accompanied, at minimum, by a certificate of analysis stating the content of the disks, lot number, expiration date, and assurance that they were tested and performed according to established QC specifications.

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3.3.2 Storing Antimicrobial Disks

Cartridges containing commercially prepared paper disks specifically for AST are generally packaged to ensure appropriate anhydrous conditions. Steps for storing and handling disks are listed in the following table.

Step	Action	Comment
1	Store disks at –20°C ± 8°C. Some disks (eg, β-lactams) should preferably be stored at –20°C.	Do not store disks in a self-defrosting freezer. Sealed packages of disks that contain drugs from the β-lactam class should be stored frozen, except for a small working supply, which may be refrigerated for 1 week at most. Some labile agents (eg, imipenem, cefaclor, and clavulanate combinations) may retain greater stability if stored frozen until the day of use.
2	Remove the sealed packages containing disk cartridges from the refrigerator or freezer 1–2 hours before use so they may equilibrate to room temperature before opening.	Equilibrating the packages to room temperature minimizes the amount of condensation that occurs when warm air makes contact with cold disks.
3	Once a cartridge of disks has been removed from its sealed package, place it in a tightly sealed, desiccated container for storage. When a disk-dispensing apparatus is used, it should be stored in a container fitted with a tight cover and supplied with an adequate desiccant.	Let the dispenser warm to room temperature before opening. Avoid excessive moisture by replacing the desiccant when the indicator changes color.
4	Refrigerate the container with the disk-dispensing apparatus when not in use.	
5	Use only disks that have not reached the manufacturer's expiration date stated on the label.	Discard disks when they reach the expiration date.

3.4 Preparing Inoculum for Disk Diffusion Tests

3.4.1 Turbidity Standard for Inoculum Preparation

To standardize the inoculum density for an AST, a $BaSO_4$ turbidity standard equivalent to a 0.5 McFarland standard or its optical equivalent (eg, latex particle suspension) should be used. A $BaSO_4$ 0.5 McFarland standard should be prepared as described in Appendix A, section A2. Alternatively, a photometric device or nephelometer along with the manufacturer's recommended standards can be used.

3.4.2 Colony Suspension Method for Inoculum Preparation

The colony suspension method (formerly direct colony suspension method) is the most convenient method for inoculum preparation. This method can be used with most organisms and is recommended for testing the fastidious organisms *H. influenzae, H. parainfluenzae, N. gonorrhoeae, N. meningitidis,* and streptococci (see Subchapter 3.8) and for testing staphylococci to detect methicillin (oxacillin) resistance. The steps for preparing inoculum by the colony suspension method are listed in the following table.

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Step	Action	Comment
1	Make a direct suspension of well-isolated colonies of the same morphological type selected from an 18- to 24-hour agar plate. The suspension should be made in saline, tryptic soy broth, or other broth that is not antagonistic to the antimicrobial agents tested. For *N. gonorrhoeae*, prepare the suspension in MHB or 0.9% phosphate-buffered saline, pH 7.	Use a nonselective medium, such as blood agar. For *Haemophilus* spp., *N. gonorrhoeae*, and *N. meningitidis*, colonies should be picked from chocolate agar. For *Streptococcus* spp., use sheep blood agar.
2	Adjust the suspension to achieve a turbidity equivalent to a 0.5 McFarland standard. This step results in a suspension containing approximately 1–2 × 10^8 CFU/mL for *E. coli* ATCC® 25922. The inoculum should be used within 15 minutes.	Use a calibrated photometric device or nephelometer. If the suspension adjustment is performed visually, use adequate light to compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines (eg, Wickerham card[19]).

Abbreviations: ATCC®, American Type Culture Collection; CFU, colony-forming unit(s); MHB, Mueller-Hinton broth; pH, negative logarithm of hydrogen ion concentration.

3.4.3 Broth Culture Method for Inoculum Preparation

Alternatively, the broth culture method (formerly growth method) can be used and is sometimes preferable when colony growth is difficult to suspend directly, and a smooth suspension cannot be made. It can also be used for nonfastidious organisms (except staphylococci) when fresh (18- to 24-hour) colonies are not available. The steps for preparing the inoculum by the broth culture method are listed in the following table.

Step	Action	Comment
1	Select at least 3–5 well-isolated colonies of the same morphological type from an agar plate culture.	
2	Touch the top of each colony with a loop or sterile swab and transfer the growth into a tube containing 4–5 mL of a broth medium, such as tryptic soy broth or other broth that is not antagonistic to the antimicrobial agents tested.	
3	Incubate the broth culture at 35°C ± 2°C until it achieves or exceeds the turbidity of the 0.5 McFarland standard (usually 2–6 hours).	
4	Adjust the turbidity of the actively growing broth culture with sterile saline or broth to achieve a turbidity equivalent to that of a 0.5 McFarland standard. This step results in a suspension containing approximately 1–2 × 10^8 CFU/mL for *E. coli* ATCC® 25922. The inoculum should be used within 15 minutes.	• Use a calibrated photometric device or nephelometer. If the suspension adjustment is performed visually, use adequate light to compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines (eg, Wickerham card[19]). • **NOTE:** Avoid extremes in inoculum density that might arise from prolonged growth because this can affect organism viability. Never use undiluted overnight broth cultures or other unstandardized inocula for streaking plates.

Abbreviations: ATCC®, American Type Culture Collection; CFU, colony-forming unit(s).

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3.4.4 Performing Colony Counts

Laboratories are encouraged, although not required, to perform colony counts on inoculum suspensions at least quarterly to ensure the final inoculum concentration routinely obtained closely approximates 1 to 2 × 10^8 colony-forming units (CFU)/mL for *E. coli* ATCC® 25922. Laboratories are also encouraged to sample the growth control for each isolate tested to ensure purity and sufficient inoculum and provide fresh colonies for repeat or additional testing if needed, without delay.

The steps for performing colony counts are listed in the following table.

Step	Action	Comment
1	Remove a 0.01-mL aliquot from the 0.5 McFarland suspension and dilute it in 10 mL saline.	This 1:1000 dilution results in a 10^5 CFU/mL suspension.
2	Mix thoroughly.	
3	Remove 0.1 mL of the suspension created in step 1 from the tube and dilute it in 10 mL saline.	This 1:100 dilution results in a 10^3 CFU/mL suspension.
4	Mix thoroughly.	
5	Spread a 0.1-mL aliquot over the surface of a suitable agar medium.	This 1:10 dilution results in a 10^2 CFU/mL suspension or approximately 150 colonies per plate.
6	Incubate the media.	After incubation, the presence of approximately 150 colonies indicates an inoculum density of 1.5 × 10^8 CFU/mL or 1–2 × 10^8 CFU/mL.

Abbreviation: CFU, colony-forming unit(s).

3.5 Inoculating the Agar Plates

The steps for inoculating the agar plates are listed in the following table.

Step	Action	Comment
1a	Dip a sterile cotton swab into the 0.5 McFarland adjusted suspension.	Use the inoculum suspension within 15 minutes after adjusting the turbidity.
1b	Rotate the swab several times and press firmly on the inside wall of the tube above the fluid level.	Rotating the swab in this manner removes excess fluid from the swab.
2a	Inoculate the dried surface of an agar plate[a] by streaking the swab over the entire agar surface.	Inoculate all plates, regardless of plate size, with the adjusted 0.5 McFarland organism suspension.
2b	Repeat the procedure performed in step 2a by streaking 2 more times, rotating the plate approximately 60 degrees each time.	Ensures an even distribution of inoculum.
2c	Swab the rim of the agar plate.	Further ensures an even distribution of inoculum.
3	Leave the lid ajar for (ideally) 3–5 minutes but no more than 15 minutes.	Leaving the lid ajar allows any excess surface moisture to be absorbed before applying the drug-impregnated disks.

Abbreviations: HTM, *Haemophilus* test medium; MHA, Mueller-Hinton agar; MH-F, Mueller-Hinton fastidious.
[a] Agar plates can be MHA, MH-F agar, MHA with 5% sheep blood, GC, or HTM.

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3.6 Applying Disks to Inoculated Agar Plates

The steps for applying the antimicrobial disks to the inoculated agar plates are listed in the following table.

Step	Action	Comment
1a	Dispense the predetermined battery of antimicrobial disks onto the surface of the inoculated agar plate.	• Place disks that produce predictably small zones (eg, gentamicin, vancomycin) next to those that produce larger zones (eg, cephalosporins, penicillins) to avoid overlapping zones. • Whether the disks are placed individually or with a dispensing apparatus, they must be distributed evenly so they are no closer than 24 mm from center to center. • No more than 12 disks should be placed on a 150-mm plate and no more than 6 disks on a 100-mm plate, although some organisms need fewer (see Subchapter 3.8 and Appendix B).
1b	Press each disk down to ensure complete contact with the agar surface.	• Pay attention to how close the disks are to the edge of the plate, no matter how many disks are dispensed. If disks are placed too close to the edge of the plate, the zones may not be fully round with some drugs. • Because some of the drug diffuses almost instantaneously, a disk should not be relocated once it has made contact with the agar surface. Instead, place a new disk in another location on the agar.
1c	If performing a test for inducible clindamycin resistance, see Subchapter 3.10.1 and CLSI M100[1] Tables 3 for guidance on disk placement.	Erythromycin and clindamycin disks must be placed at a specific distance from each other.
2a	Invert the plates and place them in an incubator set to 35°C ± 2°C within 15 minutes after the disks are applied.	Testing at temperatures above 35°C may not detect MRS.
2b	Do not incubate the plates in an atmosphere with increased CO_2 (except for *H. influenzae, H. parainfluenzae, N. gonorrhoeae, N. meningitidis,* and streptococci [see Subchapter 3.8]).	The breakpoints were developed using ambient air incubation, and CO_2 significantly alters the size of the inhibitory zones of some agents (see CLSI M100[1] Tables 4).

Abbreviation: MRS, methicillin (oxacillin)-resistant staphylococci.

3.7 Reading Plates and Interpreting Results

Each plate should be examined after incubating to ensure an appropriate incubation (see Subchapter 3.8 and Appendix B). If the plate was satisfactorily streaked and the inoculum concentration correct, the resulting zones of inhibition will be uniformly circular and there will be a confluent lawn of growth. If individual colonies are apparent, instead of a confluent lawn of growth, the inoculum concentration was too light, and the test must be repeated. Refer to CLSI *M02 Disk Diffusion Reading Guide*[15] for detailed guidance on reading zones of inhibition.

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The diameters of the zones of complete inhibition (as judged by the unaided eye), including the diameter of the disk, must be measured to the nearest whole millimeter using sliding calipers or a ruler held on the back of the inverted Petri plate that is held a few inches (eg, 1 to 3 inches) above a black, nonreflecting background illuminated with reflected light. Exceptions include:

- If MHA with 5% sheep blood is used (as with streptococci), the zones must be measured from the upper surface of the agar illuminated with reflected light and with the cover removed.
- For staphylococci other than *Staphylococcus aureus* (SOSA), including but not limited to *S. lugdunensis, S. pseudintermedius,* or *S. schleiferi* with oxacillin, 24 hours of incubation are needed before reporting as susceptible. Results may be reported after 18 hours, if resistant. Results from other agents should be read and reported at 16 to 18 hours.
- If testing vancomycin against *Enterococcus* spp., 24 hours of incubation are needed before reporting as susceptible. Results from other agents should be read and reported at 16 to 18 hours. For vancomycin against *Enterococcus* spp., the zone diameters need to be read with transmitted light.

The zone margin should be considered the area showing no obvious, visible growth that can be detected with the unaided eye. Faint growth of tiny colonies that can be detected only with a magnifying lens at the edge of the zone of inhibited growth should be ignored. Other situations to consider include:

- When discrete colonies grow within a clear zone of inhibition, the test should be repeated with a pure culture or subculture of a single colony from the primary culture plate. If discrete colonies continue to grow within the zone of inhibition, the colony-free inner zone should be measured.
- *Proteus* spp. strains may swarm into areas of inhibited growth around certain antimicrobial agents. For *Proteus* spp., the thin veil of swarming growth in an otherwise obvious zone of inhibition should be ignored.
- When blood-supplemented medium for testing streptococci is used, the zone of growth inhibition, not the zone of inhibition of hemolysis, should be measured.
- For trimethoprim and the sulfonamides, antagonists in the medium may allow some slight growth; therefore, slight growth (20% or less of the lawn of growth) should be disregarded, and the more obvious margin measured to determine the zone diameter.
- For *S. aureus* strains with penicillin zones ≥ 29 mm, see Subchapter 3.9.1 and CLSI M100[1] Tables 3 for information on interpreting the appearance of the zone edge (the penicillin zone-edge test for β-lactamase production).

The zone of inhibition sizes should be interpreted by referring to CLSI M100[1] Tables 2, and the organisms should be reported as susceptible, SDD, intermediate, or resistant to the agents that have been tested (see Subchapter 1.4.1 for definitions of these interpretive categories). Some agents may be reported only as susceptible or nonsusceptible, because only susceptible breakpoints are given, as no or very few resistant strains have been identified.

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3.8 Special Considerations for Fastidious Organisms

MHA, previously described for rapidly growing aerobic pathogens, is not adequate for AST of fastidious organisms (see Table 5). Disk diffusion tests are described in this standard for the following organisms:

- *H. influenzae* and *H. parainfluenzae*, using HTM; MH-F agar can be used for testing *H. influenzae* with selected antimicrobial agents
- *N. gonorrhoeae*, using GC agar base medium + 1% defined growth supplement
- *N. meningitidis*, using MHA supplemented with 5% sheep blood
- Streptococci, using MHA supplemented with 5% sheep blood; MH-F agar can be used for testing *Streptococcus pneumoniae*

Details of the conditions needed for testing these organisms are included in Table 5 and summarized in Appendix B. Other fastidious bacteria may be tested by a dilution or disk diffusion method as described in CLSI M45.[5] Methods for testing anaerobic bacteria are provided in CLSI M11.[3]

Table 5. Testing Considerations for Fastidious Organisms

Step	Action	Organisms (see NOTES for specific organisms below table)			
		H. influenzae* and *H. parainfluenzae	***N. gonorrhoeae***	***N. meningitidis* (refer to NOTE 2 before testing)**	***S. pneumoniae* and Other *Streptococcus* spp.**
1	Using the colony suspension procedure (see Subchapter 3.4.2), prepare a suspension in broth or saline from organisms grown and incubated on the source plates indicated in the organism-specific columns in this step. Adjust the suspension with broth or saline to achieve a turbidity equivalent to a 0.5 McFarland standard.	• Incubate the chocolate agar plate for 20–24 hours in 5% CO_2. • Adjust the suspension using a calibrated photometric device, using care in preparation (the suspension will contain approximately 1–4 × 10^8 CFU/mL), because higher inoculum concentrations may lead to false-resistant results with some β-lactam antimicrobial agents, particularly when β-lactamase–producing strains of *H. influenzae* are tested.	• Incubate the chocolate agar plate for 20–24 hours in 5% CO_2. • Prepare 0.5 McFarland suspension in MHB or 0.9% phosphate-buffered saline, pH 7.	Incubate the chocolate agar plate for 20–24 hours in 5% CO_2.	Incubate the sheep blood agar plate for 18–20 hours (no longer) in 5% CO_2 as necessary to obtain adequate growth.

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Table 5. (Continued)

Step	Action	Organisms (see NOTES for specific organisms below table)			
		H. influenzae and *H. parainfluenzae*	*N. gonorrhoeae*	*N. meningitidis* (refer to NOTE 2 before testing)	*S. pneumoniae* and Other *Streptococcus* spp.
2	Inoculate the plates prepared with the recommended medium within 15 minutes after adjusting the turbidity of the inoculum suspension (see Appendix A for medium preparation or obtain commercially).	• Use HTM or MH-F as described below. • The HTM method has been validated for *H. influenzae* and *H. parainfluenzae* only. MH-F has been validated for *H. influenzae* only. • Chocolate MHA is not recommended for routine testing of *Haemophilus* spp.	• Use GC agar, to which a 1% defined growth supplement is added after autoclaving. • Cysteine-free growth supplement is not needed for disk testing as it is for certain agents with dilution testing (see CLSI M07[2]). • Enriched chocolate agar is not recommended for susceptibility testing of *N. gonorrhoeae*.	• Use MHA supplemented with 5% sheep blood. • Enriched chocolate agar is not recommended for susceptibility testing of *N. meningitidis* except as a growth medium for inoculum preparation.	Use MHA supplemented with 5% sheep blood or MH-F. MH-F has been validated for *S. pneumoniae* only.
3	• Dispense antimicrobial disks onto the surface of the inoculated agar plate. • See CLSI M100[1] Tables 1 or appropriate portion of CLSI M100[1] Tables 2 for agents recommended for routine testing and reporting.	• 150-mm plate: ≤ 9 disks • 100-mm plate: ≤ 4 disks	• 150-mm plate: ≤ 9 disks • 100-mm plate: ≤ 4 disks • For some agents (eg, fluoroquinolones, cephalosporins) that produce extremely large zones, only 2–3 disks may be tested per 100-mm plate.	• 150-mm plate: ≤ 5 disks • 100-mm plate: ≤ 2 disks	• 150-mm plate: ≤ 9 disks • 100-mm plate: ≤ 4 disks
4	Invert the plates and incubate, using the time and conditions shown.	• 35°C ± 2°C • 5% CO_2 • 16–18 hours	• 36°C ± 1°C (do not exceed 37°C) • 5% CO_2 • 20–24 hours	• 35°C ± 2°C • 5% CO_2 • 20–24 hours	• 35°C ± 2°C • 5% CO_2 • 20–24 hours
5	Read the zones of inhibition (see Subchapter 3.7 and CLSI *M02 Disk Diffusion Reading Guide*[15]) and interpret results using specific zone diameter breakpoints in CLSI M100.[1]	See CLSI M100[1] Tables 2.			

Abbreviations: AST, antimicrobial susceptibility testing; BSC, biological safety cabinet; BSL-2, biosafety level 2; BSL-3, biosafety level 3; CFU, colony-forming unit(s); CSF, cerebrospinal fluid; FDA, US Food and Drug Administration; HTM, *Haemophilus* test medium; MHA, Mueller-Hinton agar; MHB, Mueller-Hinton broth; MH-F, Mueller-Hinton fastidious; MIC, minimal inhibitory concentration; pH, negative logarithm of hydrogen ion concentration.

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Table 5 NOTEs (for specific organisms):

NOTE 1 for *N. gonorrhoeae:*

- *N. gonorrhoeae* with 10-unit penicillin disk zone diameters of ≤ 19 mm are likely to be β-lactamase–producing strains. However, the β-lactamase test remains preferable to other susceptibility methods for rapid, accurate recognition of this plasmid-mediated penicillin resistance (see Subchapter 3.10.2).
- Culture and AST (agar dilution and disk diffusion methods) of *N. gonorrhoeae* should be considered in cases of suspected or documented treatment failure. Antimicrobial agents recommended for routine testing include, at minimum, azithromycin, ceftriaxone, cefixime, ciprofloxacin, and tetracycline.
- For azithromycin, currently there is only a susceptible breakpoint. This breakpoint presumes that azithromycin is used in an approved regimen that includes an additional antimicrobial agent.

NOTE 2 for *N. meningitidis:*

- **Recommended precautions:** Perform all AST of *N. meningitidis* in a BSC.[20-22] Manipulating *N. meningitidis* outside a BSC is associated with increased risk for contracting meningococcal disease. Laboratory-acquired meningococcal disease is associated with a case fatality rate of 50%.[23] Exposure to droplets or aerosols of *N. meningitidis* is the most likely risk for laboratory-acquired infection. Rigorous protection from droplets or aerosols is mandated when microbiological procedures (including AST) are performed on all *N. meningitidis* isolates.[22]
- If a BSC is unavailable, manipulation of these isolates should be minimized, limited to Gram staining or serogroup identification using phenolized saline solution, while wearing a laboratory coat and gloves and working behind a full-face splash shield. BSL-3 practices, procedures, and containment equipment for activities must be used with a high potential for droplet or aerosol production and for activities involving production quantities or high concentrations of infectious materials. If BSL-2 or BSL-3 facilities are not available, isolates should be forwarded to a referral or public health laboratory with a minimum of BSL-2 facilities.
- Laboratorians who are routinely exposed to potential aerosols of *N. meningitidis* should consider vaccination according to the current recommendations of the Centers for Disease Control and Prevention Advisory Committee on Immunization Practices (http://www.cdc.gov/vaccines/acip/index.html).
- Susceptibility testing with disk diffusion of *N. meningitidis* has been validated for detecting possible emerging resistance with some antimicrobial agents. To date, resistance has mostly been found in older agents used for therapy (penicillin or ampicillin) or agents used for prophylaxis of case contacts (rifampin, ciprofloxacin). Because resistance to antimicrobial agents such as ceftriaxone or cefotaxime that are often used for therapy of invasive diseases has not been confirmed, routine testing of isolates by medical laboratories is not necessary. Prophylaxis of close contacts should not be delayed while awaiting susceptibility testing results. Ceftriaxone and cefotaxime nonsusceptible isolates are rare. If testing is performed, any isolate found to be nonsusceptible should be confirmed and submitted to a public health laboratory.

NOTE 3 for *S. pneumoniae:*

- For nonmeningitis isolates of *S. pneumoniae*, oxacillin zone sizes of ≥ 20 mm indicate susceptibility to penicillin (oral or parenteral) and other β-lactam agents (see CLSI M100[1] Tables 2); both meningitis- and nonmeningitis-susceptible interpretations may be reported for penicillin and other β-lactam agents.

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- Because zones of ≤ 19 mm with the oxacillin disk screening test occur with penicillin-resistant, intermediate, and certain susceptible strains, a penicillin and cefotaxime, ceftriaxone, or meropenem MIC should be determined on isolates of *S. pneumoniae* for which the oxacillin zones are ≤ 19 mm.[24] Isolates with oxacillin zones of ≤ 19 mm must not be reported as resistant without performing a penicillin MIC.
- For *S. pneumoniae* isolated from CSF, penicillin and cefotaxime, ceftriaxone, or meropenem should be tested by a reliable MIC method (such as that described in CLSI M07[2]) and reported routinely. Such isolates can also be tested against vancomycin using the MIC or disk diffusion method.
- Inducible clindamycin resistance can be identified in *S. pneumoniae* using the method described in Subchapter 3.10.1.

NOTE 4 for β-hemolytic streptococci:

- Penicillin and ampicillin are drugs of choice for treatment of β-hemolytic streptococcal infections. Susceptibility testing of penicillins and other β-lactams approved by the FDA for treatment of β-hemolytic streptococcal infections does not need to be performed routinely because nonsusceptible isolates (ie, penicillin MICs > 0.12 and ampicillin MICs > 0.25 µg/mL) are extremely rare in any β-hemolytic streptococcus and have not been reported for *S. pyogenes*. If testing is performed, any β-hemolytic streptococcal isolate found to be nonsusceptible should be re-identified, retested, and, if confirmed, submitted to a public health laboratory.
- Inducible clindamycin resistance can be identified in β-hemolytic streptococci using the method described in Subchapter 3.10.1.

NOTE 5 for viridans group streptococci:

- A penicillin MIC should be determined on isolates of viridans group streptococci from normally sterile body sites (eg, CSF, blood, bone). Penicillin and ampicillin disk diffusion tests are not reliable with viridans group streptococci. Oxacillin disk testing to determine penicillin susceptibility of viridans group streptococci is not recommended.

3.9 Special Considerations for Detecting Resistance

3.9.1 Staphylococci

3.9.1.1 Penicillin Resistance and β-Lactamase

Most staphylococci are resistant to penicillin, which is rarely an option for treating staphylococcal infections. Penicillin-resistant staphylococcal strains produce β-lactamase, and testing penicillin instead of ampicillin is preferred. Penicillin should be used to test the susceptibility of all staphylococci to all penicillinase-labile penicillins (see CLSI M100[1] glossaries).

Some β-lactamase–producing staphylococcal isolates test susceptible to penicillin. Because staphylococcal β-lactamase is readily inducible, there is a risk of this occurring if penicillin were used to treat such strains. For this reason, it is recommended that *Staphylococcus* spp. isolates with penicillin MICs ≤ 0.12 µg/mL or zone diameters ≥ 29 mm be tested for β-lactamase production before the isolate is reported as penicillin susceptible. Several tests for β-lactamase production have been described. These tests include nitrocefin-based tests or evaluating the zone edge of a penicillin disk diffusion zone-edge test (ie, a fuzzy zone edge indicates no β-lactamase production, whereas a sharp edge indicates β-lactamase production for *S. aureus* that have penicillin zones ≥ 29 mm).[25]

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Recommended methods for detecting penicillin resistance and β-lactamase by organism group include:

- *S. aureus:*
  - The penicillin disk diffusion zone-edge test is the most sensitive test for detecting β-lactamase production in *S. aureus* and is the recommended phenotypic test to confirm the absence of β-lactamase production and penicillin susceptibility in this species. If disk diffusion testing is done routinely for *S. aureus* and a penicillin disk is tested, no additional steps are needed. A nitrocefin-based test can be performed on *S. aureus* and, if positive, the isolate can be reported as penicillin resistant. However, if the nitrocefin test is negative, a penicillin disk diffusion zone-edge test must be performed before reporting the isolates as penicillin susceptible.
- SOSA:
  - For SOSA, including *S. lugdunensis*, only nitrocefin-based tests are recommended.
  - For the most current recommendations to detect β-lactamases in staphylococcal species, see CLSI M100[1] Tables 3.

3.9.1.2 Methicillin (Oxacillin) Resistance

Historically, resistance to the antistaphylococcal, penicillinase-stable penicillins has been referred to as "methicillin resistance," and the abbreviations "MRSA" (methicillin [oxacillin]-resistant *Staphylococcus aureus*) or "MRS" (methicillin [oxacillin]-resistant staphylococci) are still commonly used, even though methicillin is no longer available for treatment. In this standard, resistance to these agents may be referred to using several terms (eg, "MRS," "methicillin resistance," or "oxacillin resistance"). Most resistance to oxacillin in staphylococci is mediated by the *mecA* or *mecC* gene, which directs the production of supplemental penicillin-binding proteins (PBPs) called penicillin-binding protein 2a (PBP2a) or penicillin-binding protein 2c (PBP2c), respectively, during bacterial cell replication. Resistance is expressed either homogeneously or heterogeneously. Homogeneous expression of resistance is easily detected with standard testing methods, because nearly all bacterial cell progeny express the resistance phenotype. Heterogeneous expression may be more difficult to detect, because only a fraction of the progeny population (eg, 1 in 100 000 cells) expresses resistance. MICs for strains of *S. aureus* with *mecC* are typically in the resistant range for cefoxitin and/or oxacillin. *mecC* resistance cannot be detected by tests only directed at *mecA* or PBP2a.

3.9.1.2.1 Methods for Detecting Oxacillin Resistance

Methods and testing conditions recommended for detecting oxacillin resistance in staphylococci are provided in Tables 2 and 3. Oxacillin disk diffusion methods should not be used except in the case of *Staphylococcus epidermidis, Staphylococcus pseudintermedius,* and *Staphylococcus schleiferi*. Cefoxitin disk diffusion is tested as a surrogate for oxacillin for all staphylococci except for *S. pseudintermedius* and *S. schleiferi*, for which oxacillin disk diffusion should be used. For the most current recommendations regarding testing and reporting, refer to CLSI M100[1] Tables 2. Recommended methods for detecting oxacillin resistance by organism group include:

- ***S. aureus* and *S. lugdunensis:***
  - For *S. lugdunensis*, oxacillin-susceptible, *mecA*-negative strains exhibit oxacillin MICs in the range of 0.25 to 1 µg/mL, whereas *mecA*-positive strains usually exhibit MICs ≥ 4 µg/mL, characteristics more like *S. aureus* than other SOSA. Therefore, *mecA*-mediated resistance in *S. lugdunensis* is detected more accurately using *S. aureus* rather than SOSA breakpoints.
  - Cefoxitin disk diffusion and MIC tests are equivalent to oxacillin MIC tests in sensitivity and specificity for *S. aureus* and *S. lugdunensis* and are better predictors of *mecA*-mediated resistance than oxacillin agar-based methods, including the oxacillin salt agar screening plate. Therefore, cefoxitin can be used as a surrogate for oxacillin MIC tests.

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- ***S. pseudintermedius* and *S. schleiferi*:**
  - For *S. pseudintermedius* and *S. schleiferi,* methicillin (oxacillin) resistance is best determined with oxacillin MIC or oxacillin disk diffusion tests. Neither cefoxitin MIC nor cefoxitin disk tests are reliable for detecting *mecA*-mediated resistance in *S. pseudintermedius* and *S. schleiferi* (see Table 6).
- **SOSA (excluding the species listed above):**
  - For SOSA, the cefoxitin disk diffusion test has sensitivity equivalent to oxacillin MIC tests but greater specificity (ie, the cefoxitin disk test more accurately identifies oxacillin-susceptible strains than the oxacillin MIC test) (see Table 6).

Table 6. Methods or Targets for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.

Organism	Disk Diffusion		MIC		*mecA*	PBP2a	Oxacillin Salt Agar
	Cefoxitin	Oxacillin	Cefoxitin	Oxacillin			
S. aureus	Yes (16–18 h)	No	Yes (16–20 h)	Yes (24 h)	Yes	Yes	Yes (24 h)
S. lugdunensis	Yes (16–18 h)	No	Yes (16–20 h)	Yes (24 h)	Yes	Yes	No
S. epidermidis	Yes (24 h)	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
S. pseudintermedius	No	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
S. schleiferi	No	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
Staphylococcus spp. (not listed above or not identified to the species level)	Yes, with exceptions[a] (24 h)	No	No	Yes (24 h)	Yes	Yes	No

Abbreviations: h, hour(s); MIC, minimal inhibitory concentration; PBP2a, penicillin-binding protein 2a.
[a] The cefoxitin disk diffusion test may not perform reliably for all species (eg, *S. haemolyticus*) that fall into the category of *Staphylococcus* spp. (not listed above or not identified to the species level).[26]

The test condition recommendations in Table 7 improve *mecA*-mediated oxacillin resistance detection in staphylococci, especially heteroresistant MRSA.

Table 7. Procedural Recommendations for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.

Recommendation Type	Comments
General	• Prepare the inoculum using the colony suspension method (see Subchapter 3.4.2). • Incubate at temperatures between 33 to 35°C. Testing at temperatures above 35°C may not detect resistance when oxacillin-based tests are used.
Oxacillin-based tests	• Add NaCl (2% w/v; 0.34 mol/L) to the medium for both agar and broth dilution oxacillin testing but not for agar medium used for disk diffusion testing. • Incubate oxacillin MIC tests for a full 24 hours before reporting as susceptible. A resistant result can be reported earlier.
Cefoxitin-based tests	• Incubate cefoxitin disk tests for 16–18 hours for *S. aureus* and *S. lugdunensis* and 24 hours for appropriate SOSA. • Read the zone of inhibition around the cefoxitin disk using reflected light for all *Staphylococcus* spp.

Abbreviations: MIC, minimal inhibitory concentration; SOSA, staphylococci other than *Staphylococcus aureus*.

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3.9.1.2.2 Molecular Detection Methods

Tests for detecting the *mecA* and/or *mecC* genes or the protein produced by *mecA*, PBP2a (also called PBP2'), are also available and are generally accurate methods for predicting resistance to oxacillin. *mecC* resistance cannot be detected by tests directed only at *mecA* or PBP2a.

3.9.1.2.3 Reporting Oxacillin Resistance

When oxacillin test results are reported, considerations include:

- Isolates that test resistant by any of the recommended methods should be reported as oxacillin resistant.
- Resistance may be reported any time growth is observed after incubating for a minimum of 16 hours.
- If cefoxitin is used as a surrogate, the isolate should be reported as oxacillin susceptible or oxacillin resistant based on the cefoxitin result.
- Oxacillin results can be applied to other penicillinase-stable penicillins.
- In most staphylococcal isolates, oxacillin resistance is mediated by *mecA* or *mecC*, encoding PBP2a and PBP2c, respectively. Isolates that test positive for *mecA* or *mecC* or PBP2a or PBP2c should be reported as oxacillin resistant.
- *S. aureus* isolates that are positive for *mecA* or *mecC* genes with molecular methods and test susceptible to cefoxitin should be investigated for mixed specimen or contamination using a purity plate. AST and molecular methods should be repeated. If the discrepancy is not resolved, the isolates should be reported as methicillin (oxacillin) resistant. See CLSI M100[1] appendixes.
- *S. aureus* isolates that are negative for *mecA* or *mecC* genes, or that do not produce PBP2a for which the phenotypic cefoxitin test is resistant, should be investigated for mixed specimen. The organism identification, AST, and molecular methods should be repeated. If the discrepancy is not resolved, the isolates should be reported as methicillin (oxacillin) resistant. See CLSI M100[1] appendixes.
- Oxacillin-resistant staphylococci are considered resistant to all penicillins, cephems (with exception of ceftaroline), β-lactam combination agents, and carbapenems (refer to CLSI M100[1] glossaries), because most cases of documented MRS infections have responded poorly to β-lactam therapy or because clinical data that demonstrate efficacy for these agents in MRS infections are not available.

Methicillin (oxacillin)-susceptible staphylococci can be considered susceptible to:

- β-Lactam combination agents (eg, amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam)
- Oral cephems (eg, cefaclor, cefdinir, cefpodoxime, cefprozil, cefuroxime, cephalexin, loracarbef)
- Parenteral cephems including first-generation cephalosporins (eg, cefazolin), second-generation cephalosporins (eg, cefamandole, cefonicid, cefuroxime), third-generation cephalosporins (eg, cefoperazone, cefotaxime, ceftizoxime, ceftriaxone), and fourth-generation cephalosporins (eg, cefepime), cephamycins (eg, ceftmetazole, cefotetan), ceftaroline, and moxalactam
- Carbapenems (eg, doripenem, ertapenem, imipenem, meropenem)

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3.9.1.3 Vancomycin Resistance in *S. aureus*

Considerations for detecting vancomycin resistance include:

- **Vancomycin-resistant *S. aureus*:**
  - The exact resistance mechanisms that result in elevated MICs are unknown, although they likely involve alterations in the cell wall and changes in several metabolic pathways. To date, most vancomycin-resistant *S. aureus* strains appear to have developed from MRSA. All these strains contained a *vanA* gene similar to that found in enterococci.[27,28]
  - These strains are reliably detected by the broth microdilution reference method and the vancomycin agar screen test (see Subchapter 3.9.1.3.2) when the tests are incubated for a full 24 hours at 35°C ± 2°C before being reported as susceptible.
  - Breakpoints for vancomycin and *S. aureus* are ≤ 2 µg/mL for susceptible, 4 to 8 µg/mL for intermediate, and ≥ 16 µg/mL for resistant. For SOSA, breakpoints are ≤ 4 µg/mL for susceptible, 8 to 16 µg/mL for intermediate, and ≥ 32 µg/mL for resistant.
- **Vancomycin-intermediate *S. aureus*:**
  - When first described, heteroresistant vancomycin-intermediate *Staphylococcus aureus* (hVISA) isolates were noted to contain subpopulations of cells (typically 1 in every 100 000 to 1 000 000 cells) for which the vancomycin MICs were 8 to 16 µg/mL (ie, in the intermediate range). Because a standard broth microdilution test uses an inoculum of 5×10^5 CFU/mL, these resistant subpopulations may go undetected, and the vancomycin MICs determined for such isolates would fall in the susceptible range (formerly between 1 and 4 µg/mL). Many physicians and microbiologists initially were skeptical that heteroresistance would result in clinical treatment failures with vancomycin because such strains were susceptible to vancomycin by the standard CLSI broth microdilution reference method. However, CLSI reviewed both clinical and laboratory data and lowered the susceptible breakpoint for vancomycin (for *S. aureus* isolates only, not SOSA) from ≤ 4 to ≤ 2 µg/mL and the resistant breakpoint from ≥ 32 to ≥ 16 µg/mL to avoid designating some heteroresistant strains susceptible and to make the breakpoints more predictive of clinical outcome. These lower breakpoints identify the hVISA strains for which the vancomycin MICs are 4 µg/mL as intermediate. Some susceptible *S. aureus* strains for which the vancomycin MICs are 1 to 2 µg/mL may still be hVISAs.
  - Determining population analysis profiles of *S. aureus* isolates is labor intensive and not suitable for routine medical laboratory application. Unfortunately, there is no standardized technique that is convenient and reliable for detecting hVISA strains. The inability of both automated commercial systems and standard reference susceptibility testing methods to detect the hVISA phenotype makes it impossible to identify all patients whose infections may be caused by hVISAs and may therefore not respond to vancomycin therapy.[29,30]

3.9.1.3.1 Methods for Detecting *S. aureus* Isolates Not Susceptible to Vancomycin

S. aureus with vancomycin MICs ≥ 8 µg/mL can be detected by either broth dilution vancomycin MIC tests or the vancomycin agar screen test. To recognize staphylococcal strains for which the vancomycin MICs are 4 µg/mL, MIC testing must be performed, and the tests incubated for a full 24 hours at 35°C ± 2°C. MIC tests should be performed to determine the susceptibility of all staphylococci isolates to vancomycin. The disk test does not differentiate vancomycin-susceptible isolates of *S. aureus* from vancomycin-intermediate isolates or differentiate among vancomycin-susceptible, -intermediate, and -resistant isolates of SOSA, all of which give similar size zones of inhibition.

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3.9.1.3.2 Vancomycin Agar Screen for *S. aureus*

The test should be performed according to the procedure described in CLSI M100[1] Tables 3 by inoculating an isolate of *S. aureus* onto brain heart infusion (BHI) vancomycin screening agar containing 6 µg/mL of vancomycin. The inoculum should be prepared from a colony suspension to obtain a 0.5 McFarland turbidity standard. The incubation temperature is 35°C ± 2°C in ambient air, and the test should be incubated for 24 hours. The agar plates should be examined carefully with transmitted light for more than one colony or light film of growth, which indicate presumptive reduced susceptibility to vancomycin. Many *S. aureus* isolates with vancomycin MICs of 4 µg/mL do not grow on this vancomycin agar screen media. If growth is found on the vancomycin screen agar, a vancomycin MIC test must be performed to establish the MIC value. Also, there are insufficient data to recommend using this agar screen test for SOSA.

3.9.1.3.3 Reporting Vancomycin for Staphylococci

Vancomycin-susceptible staphylococci should be reported following the laboratory's routine reporting protocols. Additional considerations include:

- For strains not susceptible to vancomycin (ie, those with MICs ≥ 4 µg/mL for *S. aureus* or ≥ 8 µg/mL for SOSA), preliminary results should be reported following routine reporting protocols.
- Any *S. aureus* with MICs ≥ 8 µg/mL or SOSA with MICs ≥ 32 µg/mL should be sent to a referral laboratory for confirmation. Public health authorities should be notified according to local recommendations.
- Tables 2 and appendixes in CLSI M100[1] contain the most current recommendations for testing and reporting.

3.9.1.4 Inducible Clindamycin Resistance

Inducible clindamycin resistance can be identified using the method described in CLSI M100[1] Tables 3. See Subchapter 3.10.1 for additional information on this test.

3.9.1.5 Mupirocin Resistance

High-level mupirocin resistance (ie, MICs ≥ 512 µg/mL) occurs in *S. aureus* and is associated with the presence of the plasmid-mediated *mupA* gene.[31-33] Mupirocin use has been reported to increase rates of high-level mupirocin resistance in *S. aureus*.[34] High-level mupirocin resistance can be detected using either disk diffusion or broth microdilution tests (see CLSI M100[1] Tables 3).[35]

3.9.2 Enterococci

3.9.2.1 Penicillin and/or Ampicillin Resistance

Enterococci may be resistant to penicillin and ampicillin because of production of low-affinity PBPs or, very rarely, β-lactamase. Agar or broth dilution MIC testing and disk diffusion testing accurately detect isolates with altered PBPs but do not reliably detect isolates that produce β-lactamase. β-Lactamase–producing *Enterococcus* strains are best detected by using a direct, nitrocefin-based β-lactamase test (see Subchapter 3.10.2). A β-lactamase test is the only reliable test for detecting β-lactamase–producing *Enterococcus* spp. However, because β-lactamase–positive enterococci are rare, this test does not need to be performed routinely but can be used in selected cases if there is suspicion of β-lactamase production, eg, failure of β-lactam therapy for an isolate that tests penicillin susceptible. A positive β-lactamase test result predicts resistance to penicillin and amino- and ureidopenicillins.

Enterococcus strains with ampicillin and penicillin MICs ≥ 16 µg/mL are categorized as resistant. However, enterococci with penicillin MICs ≤ 64 µg/mL or ampicillin MICs ≤ 32 µg/mL may be susceptible to synergistic killing by these penicillins in combination with gentamicin or streptomycin (in the absence of high-level resistance to gentamicin or streptomycin, see Subchapter 3.9.2.3) if high doses of penicillin or ampicillin are used. Enterococci possessing higher penicillin levels (MICs ≥ 128 µg/mL) or ampicillin (MICs ≥ 64 µg/mL) resistance may not be

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susceptible to the synergistic effect.[36,37] Physicians' requests to determine the actual penicillin or ampicillin MIC for blood and CSF *Enterococcus* isolates should be considered.

3.9.2.2 Vancomycin Resistance

Accurately detecting vancomycin-resistant enterococci (VRE) by agar or broth dilution MIC testing necessitates incubating for a full 24 hours (rather than 16 to 20 hours) and carefully examining the plates, tubes, or wells for evidence of faint growth before reporting as susceptible. With disk diffusion, zones should be examined using transmitted light. The presence of a haze or any growth within the zone of inhibition indicates resistance. Organisms with intermediate zones should be tested by an MIC method as described in CLSI M07.[2] For isolates for which the vancomycin MICs are 8 to 16 µg/mL, biochemical tests or matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (see CLSI M100[1] Tables 3 for information on additional testing and reporting). Other identification methods such as motility and pigment production should be performed to distinguish species with acquired resistance (eg, *vanA* and *vanB*) from those with intrinsic, intermediate-level resistance to vancomycin (eg, *vanC*), such as *Enterococcus gallinarum* and *Enterococcus casseliflavus,* which often grow on the vancomycin screen plate. A vancomycin agar screen test may also be used, as described in CLSI M100[1] Tables 3.

The vancomycin agar screen test can be used in addition to the dilution methods described above for detecting VRE. The test should be performed by inoculating an enterococcal isolate onto BHI-vancomycin screening agar containing 6 µg/mL of vancomycin using the method described in CLSI M100[1] Tables 3. The inoculum should be prepared from a colony suspension to obtain a 0.5 McFarland turbidity standard. The incubation temperature is 35°C ± 2°C in ambient air, and the test should be incubated for 24 hours. The vancomycin screening agar plate should be examined for more than one colony, which indicates presumptive resistance to vancomycin. Identification confirmation should be performed as described above.

3.9.2.3 High-Level Aminoglycoside Resistance

High-level resistance to gentamicin and/or streptomycin indicates that an enterococcal isolate will not be killed by the synergistic action of a penicillin or glycopeptide combined with that aminoglycoside.[36] Agar or broth high-concentration gentamicin (500 µg/mL) and streptomycin (1000 µg/mL with broth microdilution; 2000 µg/mL with agar dilution) tests or disk diffusion tests using high-concentration gentamicin (120-µg) and streptomycin (300-µg) disks can be used to screen for this type of resistance (see CLSI M100[1] Tables 3 for tests and QC recommendations). Other aminoglycosides do not need to be tested, because their activities against enterococci are not superior to gentamicin or streptomycin. Additionally, high-level resistance to both gentamicin and streptomycin implies that resistance will be found for all aminoglycosides, making testing of additional aminoglycosides unnecessary.

3.9.3 *S. pneumoniae*

Penicillin, cefotaxime, ceftriaxone, or meropenem should be tested by an MIC method and reported routinely or as part of selective reporting with CSF isolates of *S. pneumoniae*. Such isolates can also be tested against vancomycin using the MIC or disk diffusion method. Because specific breakpoints must be used depending on the infection site and the penicillin formulation used for therapy, CLSI M100[1] Tables 2 should be consulted when penicillins and third-generation cephalosporins are reported. In CLSI M100[1] Tables 2, breakpoints are listed for intravenous penicillin therapy for meningitis and nonmeningitis infections. Separate breakpoints are included for treating less severe infections with oral penicillin.

Amoxicillin, ampicillin, cefepime, cefotaxime, ceftriaxone, cefuroxime, ertapenem, imipenem, and meropenem may be used to treat pneumococcal infections. However, reliable disk diffusion susceptibility tests with these agents do not yet exist. The *in vitro* activity of these antimicrobial agents is best determined using an MIC method.

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3.9.4 Gram-Negative Bacilli

The major mechanism of resistance to β-lactam antimicrobial agents in gram-negative bacilli is β-lactamase enzyme production. Many different enzyme types have been reported. β-Lactamases may be named after:

- Primary substrates that they hydrolyze
- Biochemical properties of the β-lactamases
- Strains of bacteria from which the β-lactamases were detected
- Patient from whom a β-lactamase–producing strain was isolated[38]

β-Lactamases may be classified as molecular class A, B, C, or D enzymes, as shown in Table 8.[38]

Table 8. Enzyme Classifications for β-Lactamases[38,39]

Class	Active Site	Characteristics	Examples
A	Serine	Broad-spectrum enzymes, including most ESBLs and serine carbapenemases	• ESBLs: derivatives of TEM, SHV, CTX-M, VEB, and PER • Carbapenemases: KPC, SME, GES, IMI/NMC
B	Zinc	Metallo-β-lactamases hydrolyzing most β-lactams, including carbapenems but not monobactams	Carbapenemases: VIM, IMP, SPM, NDM, SIM
C	Serine	Broad-spectrum enzymes hydrolyzing cephalosporin subclasses III and IV[a] and cephamycins	Cephalosporinases: ACT, CMY, DHA, PDC
D	Serine	Narrow- and broad-spectrum enzymes, oxacillin-hydrolyzing enzymes	• ESBLs: OXA-11, OXA-15 • Carbapenemases: OXA-48 like, OXA-23

Abbreviations: ESBL, extended-spectrum β-lactamase; KPC, *Klebsiella pneumoniae* carbapenemase; NDM, New Delhi metallo-β-lactamase; SME, *Serratia marcescens* enzyme.
[a] Cephalosporin subclasses III and IV include cephalosporins that are sometimes referred to as third-generation or fourth-generation, respectively.

β-Lactamase enzymes in all four classes inactivate β-lactam antimicrobial agents at different rates. The genes encoding β-lactamases may be located on chromosomes and expressed with or without induction or carried on plasmids in single or multiple copies. An isolate may produce β-lactamases and possess other resistance mechanisms, such as porin mutations or efflux pumps, that restrict antimicrobial access to their active binding sites in the bacterial cell. The various β-lactam resistance mechanisms encountered in gram-negative bacteria generate a variety of antimicrobial activities expressed as a range of MIC values. The breakpoint is expected to be the MIC or zone diameter value that differentiates β-lactamase– and/or other resistance mechanism–negative strains (susceptible) from β-lactamase– and/or other resistance mechanism–positive strains (resistant). However, weak β-lactamase activity or low-level β-lactamase expression may not necessarily mean that the isolate will be refractory to β-lactam therapy. In practice, some isolates that are interpreted as susceptible produce β-lactamases that have clinically inconsequential enzyme activity. These may be extended-spectrum β-lactamase (ESBL), AmpC, or carbapenemase-type enzymes as described in Subchapters 3.9.4.1, 3.9.4.2, and 3.9.4.3, respectively.

Identifying a specific β-lactamase resistance mechanism (eg, ESBL, carbapenemases) is not necessary for determining a susceptible or resistant interpretation. However, in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, laboratories may decide to perform phenotypic or genotypic testing for β-lactamases, and the results may be used to guide therapeutic management and for infection-prevention purposes. CLSI M100[1] Tables 3 describes tests that can be used to screen for and confirm the

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presence of ESBLs in *E. coli, Klebsiella pneumoniae, Klebsiella oxytoca,* and *Proteus mirabilis*. CLSI M100[1] Tables 3 describe tests that can be used to screen for and confirm carbapenemase production in Enterobacterales and *P. aeruginosa*.

3.9.4.1 Extended-Spectrum β-Lactamases

ESBLs are class A and occasional class D enzymes that hydrolyze extended-spectrum cephalosporins and monobactams and are commonly inhibited by clavulanate and avibactam. These enzymes arise from mutations in genes for common plasmid-encoded β-lactamases (eg, TEM-1 and SHV-1), or they may be only distantly related to a native enzyme, as in the case of the CTX-M β-lactamases. ESBLs may confer resistance to penicillins, cephalosporins, and aztreonam in isolates of *K. pneumoniae, K. oxytoca,*[40] *E. coli, P. mirabilis,*[41] and other genera of the order Enterobacterales.[38] β-Lactam breakpoints are set at MIC and zone diameter values that recognize ESBL activity, in combination with other resistance mechanisms, likely to predict clinical success or failure. Most ESBL-negative gram-negative bacilli will test susceptible; however, some strains that test susceptible may contain ESBL genes producing low enzyme amounts or enzymes that have poor hydrolytic activity. These strains are correctly categorized as susceptible when the current CLSI breakpoints published in CLSI M100[1] are used.

A similar chromosomal enzyme, OXY (formerly KOXY or K1), in *K. oxytoca* acts as an extended-spectrum penicillinase, inactivating amino- and carboxypenicillins. Overproduction of OXY enzymes caused by promoter mutations results in resistance to ceftriaxone and aztreonam (but not ceftazidime), as well as resistance to sulbactam, clavulanic acid, and tazobactam. Strains producing OXY enzymes may result in a positive ESBL confirmatory test. MIC and zone diameter breakpoints correctly predict susceptibility and resistance.

3.9.4.2 AmpC Enzymes

The AmpC β-lactamases are chromosomal or plasmid-encoded enzymes.[42] Isolates that produce AmpC enzymes may have an antimicrobial susceptibility profile similar to those that produce ESBLs in that they show reduced susceptibility to penicillins, cephalosporins, and aztreonam. However, in contrast to ESBLs, AmpC β-lactamases also inactivate cephamycins (ie, bacteria expressing AmpC enzymes test resistant to cefoxitin and cefotetan). In addition, AmpC-producing strains are resistant to the older β-lactamase inhibitor combination agents; however, newer inhibitors such as avibactam, relebactam, and vaborbactam inhibit AmpC enzymes. Isolates expressing AmpC β-lactamases may test resistant to carbapenems if accompanied by a porin mutation or in combination with overexpression of specific efflux pumps.

Chromosomal AmpC β-lactamases are found in *Enterobacter* spp., *Klebsiella* (formerly *Enterobacter*) *aerogenes*, *Citrobacter* spp., *Serratia* spp., and some other gram-negative species. The chromosomal AmpC β-lactamases are usually expressed in low amounts, but the organisms can be induced to produce higher amounts of enzyme by exposure to penicillins, carbapenems, and some cephems (eg, cefoxitin). Third- and fourth-generation cephalosporins do not induce production of AmpC enzymes but can be hydrolyzed by them. Using cephalosporins also may select for stably derepressed chromosomal variants, which can emerge during therapy.[43] Derepressed variants with elevated AmpC production are resistant to most β-lactams.

Plasmid-mediated AmpC enzymes are transmissible among bacteria and can be found in isolates of *K. pneumoniae, E. coli, Proteus* spp., *Salmonella* spp., and *Shigella* spp. Plasmid-mediated AmpC β-lactamases are expressed constitutively, usually at high levels.

Citrobacter freundii complex, *Enterobacter cloacae* complex, *Hafnia alvei, K. aerogenes, Morganella morganii, Providencia* spp., *Serratia marcescens*, and *Yersinia enterocolitica* may test susceptible to ceftriaxone, cefotaxime, ceftazidime, and ceftaroline, but these agents may be ineffective against these genera within a few days after initiation of therapy due to derepression of inducible AmpC β-lactamase. The risk of AmpC derepression during therapy is moderate to high with *C. freundii* complex, *E. cloacae* complex, and *K. aerogenes* and appears to be less

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frequent with *M. morganii, Providencia* spp., and *S. marcescens*.[44] Therefore, isolates that are initially susceptible may become resistant. Testing subsequent isolates may be warranted if clinically indicated.

There is no CLSI-validated phenotypic test to confirm the presence of plasmid-encoded AmpC β-lactamases in clinical isolates. Isolates carrying both ESBLs and plasmid-encoded AmpC β-lactamases are common in some geographical regions. Using the current breakpoints for antimicrobial agents affected by these enzyme combinations is the best approach for guiding treatment of these strains.

3.9.4.3 Carbapenemases (Carbapenem-Resistant Gram-Negative Bacilli)

Carbapenemase activity has been detected in patient isolates of gram-negative bacilli with β-lactamase enzymes in classes A, B, and D. In Enterobacterales, KPC-, GES-, IMI-, and SME-type enzymes within class A, NDM-, VIM-, and IMP-type enzymes within class B, and OXA-type enzymes[45] within class D represent major families of clinical importance (see Table 9).

NDM-type and other metallo-β-lactamase enzymes need zinc for activity and are inhibited by substances that bind zinc (eg, EDTA). *Stenotrophomonas maltophilia, Bacillus anthracis,* and some strains of *Bacteroides fragilis* produce a chromosomal metallo-β-lactamase. Other metalloenzymes may be carried on plasmids and can occur in *Acinetobacter* spp., *P. aeruginosa, S. marcescens, K. pneumoniae,* and, increasingly, in other Enterobacterales.

Carbapenemase activity in Enterobacterales and *P. aeruginosa* can be confirmed using the Carba NP colorimetric microtube assay or the modified carbapenem inactivation method (mCIM) test described in CLSI M100[1] Tables 3.[46-50] Both the Carba NP and mCIM tests may detect carbapenemase production, but neither of these tests can identify the specific enzyme present. There is no CLSI-validated phenotypic test to confirm the presence of carbapenemase activity in *Acinetobacter* spp. or other taxa listed in Table 9; therefore, a molecular test is the recommended method.

Metallo-β-lactamase activity can be detected in Enterobacterales by using the EDTA-modified carbapenem inactivation method (eCIM) test in combination with the mCIM test as described in CLSI M100[1] Tables 3. The eCIM test enables differentiation of metallo-β-lactamases from serine carbapenemases in Enterobacterales isolates that are positive for mCIM. Using current breakpoints for antimicrobial agents affected by these carbapenemases is the recommended approach for providing guidance for treating infections by Enterobacterales containing both serine and metallo-β-lactamases, including KPC-, OXA-, and NDM-type enzymes. Refer to CLSI M100[1] Tables 3 for the most current recommendations for testing and reporting.

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Table 9. β-Lactamases With Carbapenemase Activity

β-Lactamase Class[a]	Found in	Examples
A	• Enterobacterales • *P. aeruginosa*	• KPC • GES • IMI/NMC • SME (*S. marcescens* only)
B	• Enterobacterales • *P. aeruginosa* • *Acinetobacter* spp. • *B. anthracis* • *Aeromonas* spp. • *S. maltophilia*	Metallo-β-lactamases inhibited by EDTA (IMP, VIM, SPM, NDM, CphA, L1)
D	• Enterobacterales • *P. aeruginosa* • *Acinetobacter* spp.	OXA-type enzymes

Abbreviations: EDTA, ethylenediaminetetraacetic acid; KPC, *Klebsiella pneumoniae* carbapenemase; NDM, New Delhi metallo-β-lactamase; SME, *Serratia marcescens* enzyme.
[a] Carbapenemases have not yet been found in class C.

3.10 Supplemental (Not Routine) Tests

A supplemental (not routine) test detects susceptibility or resistance to a drug or drug class by a method other than routine disk diffusion or broth or agar dilution MIC and does not need additional tests to confirm susceptibility or resistance. Some supplemental tests identify a specific resistance mechanism and may be required or optional for reporting specific clinical results.

3.10.1 Inducible Clindamycin Resistance

Inducible clindamycin resistance can be identified using the method described in CLSI M100[1] Tables 3.

Macrolide-resistant isolates of *S. aureus*, SOSA, *S. pneumoniae*, and β-hemolytic streptococci may express constitutive or inducible resistance to clindamycin (23S ribosomal RNA methylation encoded by the *erm* gene, also referred to as MLS_B [macrolide, lincosamide, and type B streptogramin] resistance) or may be resistant only to macrolides (efflux mechanism encoded by the *msrA* gene in staphylococci or a *mef* gene in streptococci). Infections caused by both staphylococci and streptococci with inducible clindamycin resistance may fail to respond to clindamycin therapy.[51] Inducible clindamycin resistance can be detected for all staphylococci, *S. pneumoniae*, and β-hemolytic streptococci using a disk diffusion test.

The disk diffusion test is performed by placing a 2-µg clindamycin disk either 15 to 26 mm away (for staphylococci) or 12 mm away (for streptococci) from the edge of a 15-µg erythromycin disk on either of the following media:

- An MHA plate (for staphylococci) or an MHA with 5% of sheep blood plate (for streptococci), both inoculated using the standard disk diffusion procedure
- A blood agar plate used for the inoculum purity check

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Flattening of the clindamycin zone of inhibition adjacent to the erythromycin disk (referred to as a D-zone) indicates inducible clindamycin resistance. After 16 to 18 hours of incubation (staphylococci) or 20 to 24 hours of incubation (streptococci), organisms not showing flattening of the clindamycin zone of inhibition adjacent to the erythromycin disk in an erythromycin-resistant isolate should be reported as tested (eg, susceptible or intermediate to clindamycin). Organisms showing flattening of the clindamycin zone of inhibition adjacent to the erythromycin disk after incubation have inducible clindamycin resistance. Recommendations for QC and the most current updates for testing are provided in CLSI M100[1] Tables 3 and Tables 4.

Inducible clindamycin resistance can also be detected in these organisms with the broth microdilution test using a combination of erythromycin and clindamycin placed together in the same well (see CLSI M07[2] and CLSI M100[1] Tables 3).

3.10.2 β-Lactamase Tests

3.10.2.1 Purpose

A rapid β-lactamase test may yield clinically relevant results earlier than a disk diffusion test with *Haemophilus* spp., *N. gonorrhoeae*, and some anaerobes. A β-lactamase test is the only reliable test for detecting β-lactamase–producing *Enterococcus* spp. A positive β-lactamase test result predicts:

- Resistance to penicillin, ampicillin, and amoxicillin among *Haemophilus* spp., *N. gonorrhoeae*, and some anaerobes
- Resistance to penicillin and amino- and ureidopenicillins among staphylococci and enterococci

A negative β-lactamase test result does not rule out β-lactam resistance because of other mechanisms. β-Lactamase tests must not be used for members of the Enterobacterales, *Pseudomonas* spp., and other aerobic gram-negative bacilli, because the results may not be predictive of susceptibility to the β-lactams most often used for therapy.

3.10.2.2 Selecting a β-Lactamase Test

Detecting β-lactamase in *S. aureus* is most accurately accomplished by using the penicillin disk diffusion zone-edge test (see Subchapter 3.9.1.1 and CLSI M100[1] Tables 3). Nitrocefin-based tests can be used for *S. aureus*, but negative results should be confirmed with the penicillin zone-edge test before reporting penicillin as susceptible. Nitrocefin-based tests are the preferred method for testing SOSA, enterococci, *Haemophilus* spp., *Moraxella catarrhalis,* and *N. gonorrhoeae*[52] but need enzyme induction in SOSA, including *S. lugdunensis*. Induction can be easily accomplished by testing growth from the zone margin surrounding a cefoxitin disk test. Acidimetric β-lactamase tests have generally produced acceptable results with *Haemophilus* spp., *N. gonorrhoeae*, and staphylococci. Iodometric tests may be used for testing *N. gonorrhoeae*. To ensure accurate results when these assays are performed, known positive and negative control strains should be tested at the same time as patient isolates (see the manufacturer's recommendations for commercial tests).

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3.11 Disk Diffusion Method Limitations

3.11.1 Application to Various Organisms Groups

The disk diffusion methods described in this standard are standardized for testing rapidly growing pathogens, including *Staphylococcus* spp., *Enterococcus* spp., Enterobacterales, *P. aeruginosa, Acinetobacter* spp., and *S. maltophilia,* and have been modified for testing fastidious organisms such as *H. influenzae, H. parainfluenzae, N. gonorrhoeae, N. meningitidis,* and streptococci (see CLSI M100[1] Tables 2). For organisms excluded from CLSI M100[1] Tables 2 and not covered in other CLSI documents (eg, CLSI M45[5]), studies are not yet adequate to develop reproducible, definitive standards to interpret results. These organisms may:

- Need different media.
- Need different incubation conditions.
- Show marked strain-to-strain variation in growth rate.

For these microorganisms, an infectious diseases specialist should be consulted for guidance in determining the need for AST and for interpreting results. Published reports in the medical literature and current consensus recommendations for therapy of uncommon microorganisms may obviate the need for AST of such organisms. If testing is necessary, a dilution method usually is the most appropriate testing method, and the organism may need to be submitted to a referral laboratory.

3.11.2 Warning

Dangerously misleading results can occur when certain antimicrobial agents are tested and reported as susceptible against specific organisms. These combinations include but may not be limited to:

- Aminoglycosides, first- and second-generation cephalosporins, and cephamycins against *Salmonella* and *Shigella* spp.
- Routine testing of other β-lactam agents, except penicillin, oxacillin, cefoxitin, and ceftaroline, against *Staphylococcus* spp. is not advised.
- Aminoglycosides (except for high-level resistance testing) against *Enterococcus* spp.

In addition, some bacterial species have inherent or innate antimicrobial resistance, which is reflected in wild-type antimicrobial patterns of all or almost all species representatives. Intrinsic resistance is so common that AST is unnecessary. For example, some *Citrobacter* spp. are intrinsically resistant to ampicillin. A small percentage (1% to 3%) may appear susceptible because of method variation, mutation, or low levels of resistance expression. Testing such species is unnecessary; however, if tested, the result for an antimicrobial agent–organism combination listed as having intrinsic resistance should be reported as resistant. Refer to the intrinsic resistance tables in CLSI M100[1] appendixes for a list of such species and the antimicrobial agents to which they are resistant.

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3.11.3 Development of Resistance and Testing Repeat Isolates

Isolates that are initially susceptible may become intermediate or resistant after initiating therapy. Therefore, subsequent isolates of the same species from a similar anatomical site should be tested to detect resistance that may have developed. Development of resistance can occur within as little as one to four days and has been noted most frequently in:

- *E. cloacae* complex, *C. freundii* complex, and *K. aerogenes* with third-generation cephalosporins
- *P. aeruginosa* with all antimicrobial agents
- Staphylococci with fluoroquinolones

For *S. aureus*, vancomycin-susceptible isolates may become vancomycin intermediate during the course of prolonged therapy.

In certain circumstances, the decision to perform susceptibility tests on subsequent isolates necessitates knowledge of the specific situation and the severity of the patient's condition (eg, an isolate of *E. cloacae* complex from a blood culture on a premature infant or MRSA from a patient with prolonged bacteremia). Laboratory guidelines on when to perform susceptibility testing on repeat isolates should be determined after consultation with the antimicrobial stewardship team and other relevant institutional stakeholders.

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Chapter 4

Quality Control and Quality Assurance

4 Quality Control and Quality Assurance

4.1 Quality Control Purpose

QC includes the procedures to monitor the test system to ensure accurate and reproducible results. This outcome is achieved by but not limited to testing carefully selected QC strains with known susceptibility to the antimicrobial agents tested. The goals of a QC program are to monitor:

- Precision (reproducibility) and accuracy of susceptibility test procedures
- Performance of reagents and equipment used in the tests
- Performance of persons who carry out the tests and report the results

A comprehensive QA program helps to ensure that testing materials and processes consistently provide quality results. QA includes but is not limited to:

- Monitoring
- Evaluating
- Taking corrective actions (if necessary)
- Recordkeeping
- Calibrating and maintaining equipment
- Proficiency testing
- Training
- Competence assessment
- QC

4.2 Quality Control Responsibilities

This subchapter is intended to apply to the standard reference methods; however, it may apply to certain commercially available AST systems that are based primarily or in part on methods described in CLSI M07.[2]

Manufacturers and users of AST systems have a shared responsibility to ensure products meet performance specifications. The primary purpose of QC testing performed by manufacturers (in-house reference methods or commercial methods) is to ensure that testing materials and reagents have been appropriately manufactured and that the QC strains will provide acceptable results in the users' hands. The primary purpose of QC testing performed by laboratories (users) is to ensure that current and new testing materials and reagents are performing and maintained properly and testing is performed according to established protocols. A logical division of responsibility and accountability may be described as:

- Manufacturers (in-house or commercial products) are responsible for:
  - Antimicrobial labeling
  - Antimicrobial agent potency or concentration
  - Antimicrobial stability
  - Compliance with good manufacturing practices (eg, quality management system standards)
  - Product integrity
  - Accountability and traceability to consignee

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- Laboratories (users) are responsible for:
  - Antimicrobial agent storage under the manufacturer's recommended environmental conditions (to prevent antimicrobial agent deterioration)
  - Proficiency of personnel performing tests
  - Using current CLSI standards (or manufacturers' instructions for use) and adhering to the established procedures (eg, inoculum preparation, incubation conditions, end-point determination, results interpretation)

Manufacturers should design and recommend a QC program that enables users to evaluate variables (eg, inoculum density, storage, and shipping conditions) that are most likely to adversely affect test results and to determine that the test results are accurate and reproducible when the test is performed according to established protocols.

NOTE: Laboratories should familiarize themselves with applicable regulatory and accreditation requirements for QC (see CLSI EP23™[53]).

4.3 Selecting Strains for Quality Control

Each QC strain should be obtained from a recognized source (eg, ATCC®). CLSI-recommended QC strains appropriate for the antimicrobial agent and reference method should produce results within the expected ranges listed in CLSI M100[1] that were established according to the procedures described in CLSI M23.[4] If specialty media is required for testing, additional QC organisms to ensure accuracy of the test results may be warranted. It is the manufacturer's responsibility to ensure their products pass QC. Commercial system users should follow the QC recommendations in that system's instructions for use.

QC strains and their characteristics are described in CLSI M100[1] appendixes. Refer to CLSI M100[1] Tables 4 for recommendations for routine QC. Routine QC strains are tested regularly (eg, daily or weekly) to ensure the test system is working and produces results that fall within the specified limits in CLSI M100.[1] The QC strains recommended in CLSI M100[1] should be included when a laboratory performs CLSI reference disk diffusion testing as described herein.

Other QC strains are considered supplemental and are used to assess particular test or test system characteristics in select situations, or they may represent alternative QC strains. For example, *H. influenzae* ATCC® 10211 is more fastidious than *H. influenzae* ATCC® 49247 or *H. influenzae* ATCC® 49766 and is used to ensure HTM can adequately support the growth of *H. influenzae* and *H. parainfluenzae* patient isolates. Supplemental QC strains may possess susceptibility or resistance characteristics specific for one or more special tests listed in CLSI M02, M07,[2] and M100.[1] They can be used to assess a new test, for training new personnel, and for competence assessment. It is not necessary to include supplemental QC strains in routine daily or weekly AST QC programs.

Expected QC ranges for routine QC strains are listed in CLSI M100[1] Tables 4. When recommended, acceptable QC ranges for supplemental QC strains are also included in CLSI M100[1] Tables 3, which describe screening and confirmatory tests for specific resistance mechanisms.

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4.4 Maintaining and Testing Quality Control Strains

Proper organism storage and maintenance is needed to ensure acceptable QC strain performance (also refer to Appendix C). Considerations include:

- For long-term storage, stock cultures should be maintained at ≤ −20°C (preferably at ≤ −60°C or in liquid nitrogen) in a suitable stabilizer (eg, 50% fetal calf serum in broth, 10% to 15% glycerol in tryptic soy broth, defibrinated sheep blood, or skim milk) or in a freeze-dried state.
- Some QC strains, particularly those with plasmid-mediated resistance (eg, *E. coli* ATCC® 35218), have been shown to lose the plasmid when stored at temperatures above −60°C or repeatedly subcultured. The footnotes in CLSI M100[1] appendixes highlight special considerations for handling certain QC strains. These considerations include testing β-lactamase–producing QC strains with an antimicrobial agent, which is known to be inactivated by the β-lactamase produced.
- Appendix C illustrates workflow for subculturing and testing QC strains.
- If an unexplained QC error occurs that might be because of a change in the organism's inherent susceptibility or resistance, a fresh stock culture of the QC strain should be obtained from an external source (see Subchapter 4.8 for additional guidance).
- It is important to always test the QC strains using the same materials and methods that are used to test patient isolates.

4.5 Batch or Lot Quality Control

Guidelines for QC of each batch, shipment, or media or antimicrobial disk lot include:

- Each new batch, shipment, or lot of MHA or supplemented MHA plates or antimicrobial disks should be tested with the appropriate QC strains before or concurrent with their first use for testing patient isolates. If zone diameter measurements obtained with the batch or lot do not fall within the acceptable ranges (see CLSI M100[1] Tables 4), corrective action should be taken. Patient results can be reported for that day if QC results are within the acceptable limits (see Subchapter 4.8 for corrective action and Subchapter 4.9 to determine whether patient test results can be reliably reported). Records to maintain include, at a minimum, the lot numbers, expiration dates, and date of use of all materials and reagents used in performing susceptibility tests. See CLSI M100[1] Tables 4 for suggested QC frequency when modifications are made to antimicrobial susceptibility test systems.
- For MHA or supplemented MHA plates prepared in-house, at least one uninoculated agar plate from each batch or lot should be incubated overnight to confirm the medium's sterility.
- For certain screening, supplemental, and surrogate tests (see CLSI M100[1] Tables 3 for details), a positive (resistant) and negative (susceptible) QC strain must be tested with each new lot or shipment of disks, agar plates used for agar dilution, or single wells or tubes used with broth dilution methods. Subsequently, weekly QC testing of the negative control (susceptible strain) is sufficient if the test is performed at least once a week and criteria for converting from daily to weekly QC testing have been met (see Subchapter 4.7.2). QC of tests with the negative control (susceptible strain) is recommended each testing day if the test is not performed routinely (ie, at least once a week).

4.6 Zone Diameter Quality Control Ranges

Acceptable zone diameter QC ranges are listed in CLSI M100[1] Tables 4. These ranges were developed following specific protocols outlined in CLSI M23.[4]

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4.7 Quality Control Testing Frequency (also refer to Appendix D and CLSI M100[1] Tables 4)

Appropriate QC strains should be tested on each day the test is performed on patient isolates. Alternatively, one of two plans can be adopted to demonstrate acceptable performance to reduce the frequency of QC tests from daily to weekly. Either plan enables weekly QC testing once satisfactory performance with daily testing of QC strains is documented (see Subchapters 4.7.2.1 and 4.7.2.2). The weekly QC testing options are not applicable when tests are performed less than once a week.

4.7.1 Daily Quality Control Testing

A laboratory can perform QC testing daily. Daily (vs weekly) QC testing must be performed on each day patient isolates are tested if tests are performed less than once a week. "Daily" QC testing or testing on "consecutive" test days means testing QC strains on each day tests are performed on patient isolates. It does not refer to calendar days.

4.7.2 Performance Criteria for Reducing Quality Control Frequency to Weekly

Two plans are available to demonstrate satisfactory performance with daily QC testing before going to weekly QC testing. These are:

- The 20- or 30-day plan
- The 15-replicate (3- × 5-day) plan

4.7.2.1 The 20- or 30-Day Plan

When the 20- or 30-day plan is implemented:

- All applicable QC strains are tested for 20 or 30 consecutive test days, and results are documented.
- Recommended actions as described in Appendix D should be followed.
- If no more than one of 20 or three of 30 zone diameter measurements for each antimicrobial agent and organism combination are outside the acceptable zone diameter QC range listed in CLSI M100[1] Tables 4, it is acceptable to go to weekly QC testing.
- If completion of the 20- or 30-day plan is unsuccessful, corrective action should be taken as appropriate and daily QC testing continued.
- If a laboratory routinely tests QC strains on each day of use and desires to convert to a weekly QC plan, it is acceptable to retrospectively analyze QC data from consecutive tests available during the previous two years, providing no aspects of the test system have changed.

4.7.2.2 The 15-Replicate (3- x 5-Day) Plan

When the 15-replicate (3- × 5-day) plan is used:

- Three replicates of each applicable QC strain are tested using individual inoculum preparations for five consecutive test days and the results are documented.
- Recommended actions as described in Appendix D and Table 10 should be followed.
- Following successful completion of the 15-replicate (3- × 5-day) plan, it is acceptable to go to weekly QC testing.
- If completion of the 15-replicate (3- × 5-day) plan is unsuccessful, corrective action should be taken, as appropriate, and daily QC testing continued.

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Table 10. 15-Replicate (3- × 5-Day) Plan: Acceptance Criteria and Recommended Action[a]

Number Out of Range With Initial Testing (based on 15 replicates)	Conclusion From Initial Testing (based on 15 replicates)	Number Out of Range After Repeat Testing (based on all 30 replicates)	Conclusion After Repeat Testing
0–1	Plan is successful. Convert to weekly QC testing.	N/A	N/A
2–3	Test another 3 replicates for 5 days.	2–3	Plan is successful. Convert to weekly QC testing.
≥ 4	Plan fails. Investigate and take corrective action as appropriate. Continue QC each test day.	≥ 4	Plan fails. Investigate and take corrective action as appropriate. Continue QC each test day.

Abbreviations: N/A, not applicable; QC, quality control.
[a] Each QC strain and antimicrobial agent combination is assessed separately.

For background information that supports the 3- × 5-day plan, refer to the Statisticians' Summary for Alternative QC Frequency Testing Proposal (available at https://clsi.org/media/1719/statisticians_summary_alternative_qc_frequency_testing_proposal.pdf).

4.7.2.3 Implementing Weekly Quality Control Testing

Weekly QC testing may be implemented once satisfactory performance with daily QC testing has been documented (see Subchapters 4.7.2.1 and 4.7.2.2).

- QC testing must be performed once per week and whenever any test reagent component (eg, a new lot of agar or a new lot of disks from the same or a different manufacturer) is changed.
- If any of the weekly QC results are out of range, corrective action should be taken.
- Refer to CLSI M100[1] Tables 4 for guidance on QC frequency with new materials or test modifications.

4.8 Out-of-Range Results With Quality Control Strains and Corrective Action

Out-of-range QC results can be categorized into random, identifiable, or system related:

- QC ranges are established to include ≥ 95% of results obtained from routine testing of QC strains. A small number of (random) out-of-range QC results may be obtained even when the test method is performed correctly, and materials are maintained according to recommended protocols. Such occurrences can be observed based on normal variability of the test.
- Out-of-range results with QC strains caused by random or identifiable errors can usually be resolved by a single repeat of the QC test. However, out-of-range QC results that are caused by a problem with the test system usually do not correct when the QC test is repeated and may indicate a serious problem that can adversely affect patient results. Every out-of-range QC result must be investigated.

NOTE: See the disk diffusion troubleshooting guide in CLSI M100[1] Tables 4 for troubleshooting and corrective action for out-of-range results with QC strains.

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4.8.1 Daily or Weekly Quality Control Testing: Out-of-Range Result Caused by Identifiable Error

If the reason for an out-of-range result can be identified and easily corrected:

1. Correct the problem.
2. Document the reason.
3. Retest the QC strain on the day the error is observed.

If the repeated result is within range, no additional corrective action is needed. Identifiable reasons for the out-of-control results should be documented and may include but are not limited to:

- QC strain:
  - Changes in the organism (eg, mutation, loss of plasmid)
  - Contamination
  - Improper storage
  - Inadequate maintenance (eg, preparing a QC plate for testing from a plate other than the F2 subculture plate that is less than seven days old)
  - Nonviability
  - Using the wrong QC strain
- Testing supplies:
  - Contamination
  - Improper storage or shipping conditions
  - Using defective (eg, too thick or too thin) agar plates
  - Using damaged (eg, cracked) plates
  - Using defective (ie, inaccurate drug potency) disks
  - Using expired materials
- Testing process:
  - Incorrectly prepared or adjusted inoculum suspensions
  - Contamination
  - Inoculum prepared from a plate incubated for incorrect time frame
  - Inoculum prepared from differential or selective media containing antimicrobial agents or other growth-inhibiting compounds
  - Using the wrong incubation temperature or conditions
  - Using an incorrect incubation time
  - Using the wrong disks and/or ancillary supplies
  - Improperly placing disks (eg, inadequate contact with the agar) or disk falls off agar
  - Incorrectly reading or interpreting test results
  - Transcription error
- Equipment:
  - Not functioning properly or out of calibration (eg, pipettes, photometric device)

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4.8.2 Daily Quality Control Testing: Out-of-Range Result Not Caused by Identifiable Error

Corrective action needs to be taken if results for a QC strain and antimicrobial agent combination are out of range and the error is not identifiable and occurs on two consecutive testing days. Corrective action also needs to be taken if more than three results for a QC strain and antimicrobial agent combination are out of range during 30 consecutive testing days (see QC ranges listed in CLSI M100[1] Tables 4).

4.8.3 Weekly Quality Control Testing: Out-of-Range Result Not Caused by Identifiable Error

If the reason for the out-of-range result with the QC strain cannot be identified, corrective action needs to be taken to determine whether the error is random or system related:

- The out-of-range antimicrobial agent and organism combination must be tested on the day the error is observed or as soon as an F2 or F3 subculture of the QC strain is available (see Appendix C).
- If the repeat results are in range, all QC results available for the antimicrobial agent and organism combination when the same lot numbers of materials that were used when the out-of-range QC result was observed are used, they need to be evaluated. If five acceptable QC results are available, no additional days of QC testing are needed.

The following tables illustrate two scenarios that might be encountered, along with suggested actions.

Scenario 1: Ampicillin *E. coli* ATCC® 25922; acceptable range, 15 to 22 mm

Week	Day	Lot Number (disks)	Lot Number (MHA)	Result, mm	Action
1	1	3564	16481	18	
2	1	3564	16481	19	
3	1	3564	16481	18	
4	1	3564	16481	19	
5	1	3564	16481	14	Out of range. Repeat QC same day.
5	2	3564	16481	17	In range. Five acceptable in-range QC tests for *E. coli* ATCC® 25922 with ampicillin disks lot 3564 and MHA lot 16481. Resume weekly QC testing.

Abbreviations: ATCC®, American Type Culture Collection; MHA, Mueller-Hinton agar; QC, quality control.

Conclusion: Random QC error.

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Scenario 2: Ampicillin *E. coli* ATCC® 25922; acceptable range, 15 to 22 mm

Week	Day	Lot Number (disks)	Lot Number (MHA)	Result, mm	Action
1	1	9661	16922	18	
2	1	9661	16922	19	
3	1	9661	16922	14	Out of range. Repeat QC same day.
3	2	9661	16922	18	In range. Three acceptable in-range QC tests for *E. coli* ATCC® 25922 with ampicillin disks lot 9661 and MHA lot 16922. Repeat QC for 2 more consecutive days.
3	3	9661	16922	18	In range.
3	4	9661	16922	17	In range. Five acceptable in-range QC tests for *E. coli* ATCC® 25922 with ampicillin disks lot 9661 and MHA lot 16922. Resume weekly QC testing.

Abbreviations: ATCC®, American Type Culture Collection; MHA, Mueller-Hinton agar; QC, quality control.

Conclusion: Random QC error.

4.8.4 Additional Corrective Action

If repeat results with QC strains are still out of range, additional corrective action is needed.

- It is possible that the problem is caused by a system error rather than a random error (see Subchapter 4.8.1 and CLSI M100[1] Tables 4).
- Daily QC tests must be continued until the problem is resolved.
- If necessary, a new QC strain (either from stock cultures or a reliable source) and new lots of materials (including new turbidity standards) need to be obtained, possibly from different manufacturers. If the problem appears to be related to a manufacturer, the manufacturer should be contacted and provided with the test results and lot numbers of materials used. It may be helpful to exchange QC strains and materials with another laboratory using the same method to determine the root cause of out-of-range QC results when the reason is not identifiable. Until the problem is resolved, it may be necessary to use an alternative test method.

4.9 Reporting Patient Results When Out-of-Range Quality Control Results Are Observed

When an out-of-range result occurs when QC strains are tested or when corrective action is necessary, each patient test result must be carefully examined to determine whether it can be reliably reported. Factors to consider may include but are not limited to:

- Size and direction of QC strain error (eg, slightly or significantly increased or decreased zone size)
- Actual patient result and its proximity to the breakpoint
- Results with other QC organisms
- Results with other antimicrobial agents
- Usefulness of the particular QC strain and antimicrobial agent as an indicator for a procedural or storage issue (eg, inoculum dependent, heat labile) (refer to CLSI M100[1] Tables 4)

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Options to consider for patient results include:

- Suppressing the results for an individual antimicrobial agent
- Reviewing individual patient or cumulative data for unusual patterns
- Using an alternative test method or a referral laboratory until the problem is resolved

4.10 Confirming Results When Patient Isolates Are Tested

Multiple test parameters are monitored by following the QC recommendations described in this standard. However, acceptable results derived from testing QC strains do not guarantee accurate results when patient isolates are tested. It is important to review all results obtained from all antimicrobial agents tested on a patient's isolate before reporting the results. This review should include ensuring that:

- AST results are consistent with the isolate's identification.
- AST results for individual antimicrobial agents within a specific drug class follow the established hierarchy of activity rules (eg, third-generation cephalosporins are more active than first- or second-generation cephalosporins against Enterobacterales).
- The isolate is susceptible to antimicrobial agents for which resistance has not been documented (eg, vancomycin and *Streptococcus* spp.) and for which only "susceptible" breakpoints exist in CLSI M100[1] Tables 2.

Unusual or inconsistent results should be confirmed by checking for:

- Previous results on the patient (eg, whether the patient previously had the same isolate with an unusual antibiogram)
- Previous QC performance (eg, whether there is a similar trend or observation with recent QC testing, such as smaller QC zone sizes with a new lot of materials and an increase in resistance with patient isolates)
- Problems with the testing supplies, process, or equipment (see Subchapter 4.8.1 and CLSI M100[1] Tables 4)

If a reason for the unusual or inconsistent result for the isolate cannot be ascertained, a repeat of the antimicrobial susceptibility test or the identification, or both, may be needed. Sometimes, it is helpful to use an alternative test method for the repeat test. A suggested list of results that may need verification is included in CLSI M100[1] appendixes. Each laboratory must develop its own policy for confirming unusual or inconsistent AST results. This policy should emphasize results that may significantly affect patient care.

4.11 End-Point Interpretation Control

End-point interpretation needs to be monitored to minimize variation in zone size end-point interpretation among observers. All laboratory personnel who perform these tests should independently read a selected set of disk diffusion tests. The results should be recorded and compared with the results obtained by an experienced reader or, when QC strains are used, compared with the expected results from CLSI M100[1] Tables 4. Generally, readings from several individuals of the same zone on the same plate should not vary more than ± 2 mm. This process may be included as a part of annual and semiannual competence assessment.

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Chapter 5

Conclusion

5 Conclusion

This standard presents a process workflow for standard disk diffusion techniques, which includes agar plate preparation, testing conditions, results interpretation, disk diffusion method limitations, and QC procedures. Using the methods in this standard, along with the corresponding CLSI M100[1] supplement, enables laboratories to assist the clinician in selecting appropriate antimicrobial therapy for patient care.

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Chapter 6

Supplemental Information

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Appendix A. Preparation of Media and Reagents

Abbreviations for Appendix A

HTM *Haemophilus* test medium

MHA Mueller-Hinton agar

MH-F Mueller-Hinton fastidious

NAD β-nicotinamide adenine dinucleotide

pH negative logarithm of hydrogen ion concentration

A1 Agar Media

A1.1 Mueller-Hinton Agar

The steps for preparing Mueller-Hinton agar (MHA) are listed in the following table.

Step	Action	Comment
1	Prepare medium according to the manufacturer's instructions.	
2	Autoclave the medium and let cool to 45 to 50°C in a water bath.	
3	Pour the freshly prepared and cooled medium into glass or plastic flat-bottomed Petri plates on a level, horizontal surface. The medium volume in the plate should be uniform and approximately 4 mm deep.	• The agar volume required to provide the specified depth is calculated from the formula 3.143 × plate radius (cm)^2 × depth (cm).[1] • This corresponds to 60–70 mL for a 150-mL plate with a diameter of 150 mm and 25–30 mL for a plate with a diameter of 100 mm.
4	Let the agar plates cool to room temperature.	The plates can be used the same day or stored in a refrigerator (2 to 8°C). Plates are stable for 7 days but could have a longer shelf life if precautions are taken to prevent drying and QC is in range at the time of testing.
5	When the medium is prepared, check the pH of each MHA batch after it has cooled and solidified.	• The method used depends on the type of equipment available. Check the pH by: – Macerating enough agar to submerge the tip of a pH electrode – Letting a small amount of agar solidify around the tip of a pH electrode in a beaker or cup – Using a surface electrode • The agar medium should have a pH between 7.2 and 7.4 at room temperature. If the pH is < 7.2, certain drugs will appear to lose potency (eg, aminoglycosides, macrolides), whereas other antimicrobial agents may appear to have excessive activity (eg, tetracyclines). If the pH is > 7.4, the opposite effects can be expected.
6	Incubate a representative sample of each batch at 35°C ± 2°C for 24 hours or longer and examine for sterility.	

Abbreviations: MHA, Mueller-Hinton agar; pH, negative logarithm of hydrogen ion concentration; QC, quality control.

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Appendix A. (Continued)

NOTE: Supplemental calcium or magnesium cations must not be added to MHA.[1]

A1.2 Mueller-Hinton Agar + 5% Sheep Blood

The steps for preparing MHA + 5% sheep blood are listed below.

1. Prepare MHA as described in section A1.1 (step 1). When MHA has cooled to 45 to 50°C, add 50 mL defibrinated sheep blood to 1 L MHA. Continue as described in section A1.1 (step 3).
2. Check the pH after the blood is aseptically added to the autoclaved and cooled medium. The final pH should be the same as that for unsupplemented MHA (7.2 to 7.4).

A1.3 GC Agar + 1% Defined Growth Supplement

The steps for preparing GC agar + 1% defined growth supplement are listed in the following table.

Step	Action	Comment
1	Prepare 1 L single-strength GC agar base from a commercially available dehydrated base according to the manufacturer's instructions.	
2	Autoclave the media.	
3	Cool the media to 45 to 50°C in a 45 to 50°C water bath.	
4	Prepare a 1% defined growth supplement containing the specified ingredients per L.	• 1.1 g L-cystine • 0.03 g guanine HCl • 0.003 g thiamine HCl • 0.013 g *para*-aminobenzoic acid • 0.01 g vitamin B_{12} • 0.1 g thiamine pyrophosphate (cocarboxylase) • 0.25 g NAD • 1 g adenine • 10 g L-glutamine • 100 g glucose • 0.02 g ferric nitrate • 25.9 g L-cysteine HCl
5	Add 10 mL 1% defined growth supplement to the cooled GC agar.	
6	Check the pH of the medium after the defined growth supplement is aseptically added to the autoclaved and cooled medium.	The pH should be 7.0–7.4.

Abbreviations: NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration.

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Appendix A. (Continued)

A1.4 *Haemophilus* Test Medium Agar

In its agar form, *Haemophilus* test medium (HTM) consists of:

- MHA
- 15 µg/mL β-nicotinamide adenine dinucleotide (NAD)
- 15 µg/mL bovine or porcine hematin
- 5 g/L yeast extract

The steps for preparing HTM agar are listed in the following table.

Step	Action	Comment
1	Prepare a fresh hematin stock solution by dissolving 50 mg hematin powder in 100 mL 0.01 mol/L NaOH with heat.	Stir the solution until the powder is thoroughly dissolved.
2	Prepare an NAD stock solution by dissolving 50 mg NAD in 10 mL distilled water. Filter to sterilize.	
3	Prepare MHA from a commercially available dehydrated base according to the manufacturer's instructions.	
4	Add 5 g yeast extract and 30 mL hematin stock solution to 1 L MHA.	
5	Autoclave the media.	
6	Cool the media to 45 to 50°C.	
7	Aseptically add 3 mL of the NAD stock solution.	
8	Check the pH of the medium after the NAD is added to the autoclaved and cooled medium.	The pH should be 7.2–7.4.

Abbreviations: ATCC®,[a] American Type Culture Collection; HTM, *Haemophilus* test medium; MHA, Mueller-Hinton agar; NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration; QC, quality control.
[a] ATCC® is a registered trademark of the American Type Culture Collection.

NOTE: *Haemophilus influenzae* (ATCC® 10211) is recommended as a useful additional QC strain to verify the growth promotion properties of HTM. In particular, HTM manufacturers are encouraged to use *H. influenzae* ATCC® 10211 as a supplemental QC test strain.

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Appendix A. (Continued)

A1.5 Mueller-Hinton Fastidious Agar

Mueller-Hinton fastidious (MH-F) agar consists of MHA supplemented with 5% mechanically defibrinated horse blood and 20 mg/L NAD.

To prepare MH-F agar for testing *H. influenzae* or *Streptococcus pneumoniae*, the steps for preparing MHA in sections A1.1 and A1.2 should be followed. After the medium is autoclaved and cooled, 1 mL of 20 mg/mL NAD and 50 mL mechanically defibrinated horse blood are added to 1 L medium before plates are poured. The final pH should be the same as for MHA (7.2 to 7.4). The steps for preparing MH-F agar are listed in the following table.

Step	Action	Comment
1	Prepare an NAD stock solution by dissolving 20 mg NAD per 1 mL distilled water. Filter to sterilize.	1 mL of a 20 mg/mL NAD solution is needed for each liter of MH-F agar. Weigh > 20 mg NAD and determine the water volume needed by dividing the NAD weight by 20 mg/mL. For example, if 45 mg NAD is weighed, divide 45 by 20, and dissolve NAD in 2.25 mL distilled water. The remaining solution can be stored at ≤ –70°C for 1 year.
2	Prepare MHA from a commercially available dehydrated base according to the manufacturer's instructions.	
3	Autoclave the MHA medium.	
4	Cool the medium to 45 to 50°C.	
5	Aseptically add 1 mL of the 20 mg/mL NAD stock solution to 1 L medium.	
6	Aseptically add 50 mL of mechanically defibrinated horse blood to 1 L medium.	
7	Check the pH of the medium after the NAD and defibrinated horse blood are added to the autoclaved and cooled medium.	The pH should be 7.2–7.4.

Abbreviations: MHA, Mueller-Hinton agar; MH-F, Mueller-Hinton fastidious; NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration.

NOTE: Mechanically defibrinated horse blood is available from commercial sources.

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Appendix A. (Continued)

A2 Reagent: 0.5 McFarland Turbidity Standard

The steps for preparing a 0.5 McFarland turbidity standard are listed in the following table.

Step	Action	Comment
1	Prepare a 0.048-mol/L $BaCl_2$ (1.175% w/v $BaCl_2 \cdot 2H_2O$) stock solution in laboratory-grade water.	
2	Prepare a 0.18-mol/L (0.36 N) H_2SO_4 (1% v/v) stock solution in laboratory-grade water.	
3	Add 0.5 mL of the $BaCl_2$ solution to 99.5 mL of the H_2SO_4 stock solution and stir constantly.	Stirring helps maintain a suspension.
4	Verify that the density of the turbidity standard is correct by measuring absorbance using a spectrophotometer with a 1-cm light path and matched cuvettes.	The absorbance at 625 nm should be 0.08–0.13 for the 0.5 McFarland standard.
5	Transfer the $BaSO_4$ suspension to tubes in 4- to 6-mL aliquots.	Use screw-cap tubes of the same size as those used for standardizing the bacterial inoculum.
6	Tightly seal the tubes and store in the dark at room temperature.	
7	Vortex the $BaSO_4$ turbidity standard vigorously before each use and inspect for a uniformly turbid appearance.	• Replace the standard if large particles appear. • **NOTE:** McFarland standards made from latex particle suspension are commercially available. When used, they should be mixed by inverting gently (not on a vortex mixer) immediately before use. • $BaSO_4$ standards should be replaced or their densities verified monthly.

Reference for Appendix A

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Appendix B. Conditions for Disk Diffusion Antimicrobial Susceptibility Tests

Conditions needed for performing disk diffusion with overnight antimicrobial susceptibility tests on nonfastidious and fastidious organisms are listed in Tables B1 and B2, respectively. For instructions for performing disk diffusion directly from blood cultures, refer to Appendix E and CLSI M100.[1]

Table B1. Conditions for Disk Diffusion Antimicrobial Susceptibility Tests for Nonfastidious Organisms (refer to CLSI M100[1] Tables 2)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal QC[a]	Comments/ Modifications
Enterobacterales (including *Salmonella* and *Shigella* spp.)	MHA	Colony suspension in broth or saline, or broth culture method	35°C ± 2°C; ambient air	16–18 hours	• *Escherichia coli* ATCC®[b] 25922 • *P. aeruginosa* ATCC® 27853 for carbapenems • *Staphylococcus aureus* ATCC® 25923 when testing azithromycin against *Salmonella* enterica ser. Typhi or *Shigella* spp. • For β-lactam combination agents, refer to CLSI M100[1] Tables 4.	
Pseudomonas aeruginosa	MHA	Colony suspension in broth or saline, or broth culture method	35°C ± 2°C; ambient air	16–18 hours	• *P. aeruginosa* ATCC® 27853 • For β-lactam combination agents, refer to CLSI M100[1] Tables 4.	

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Table B1. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal QC[a]	Comments/Modifications
***Acinetobacter* spp.**	MHA	Colony suspension in broth or saline, or broth culture method	35°C ± 2°C; ambient air	20–24 hours	• *P. aeruginosa* ATCC® 27853 • *E. coli* ATCC® 25922 for tetracyclines and trimethoprim-sulfamethoxazole • For β-lactam combination agents, refer to CLSI M100[1] Tables 4.	
Stenotrophomonas maltophilia	MHA	Colony suspension in broth or saline, or broth culture method	35°C ± 2°C; ambient air	20–24 hours	• *P. aeruginosa* ATCC® 27853 • *E. coli* ATCC® 25922 for minocycline and trimethoprim-sulfamethoxazole	
***Staphylococcus* spp.**	MHA	Colony suspension in broth or saline	35°C ± 2°C; ambient air	16–18 hours; 24 hours for cefoxitin when testing *Staphylococcus* spp., except *S. aureus, S. lugdunensis, S. pseudintermedius,* and *S. schleiferi*	*S. aureus* ATCC® 25923	• Testing at temperatures above 35°C may not detect MRS. • Organisms with resistant results by disk diffusion should be confirmed using an MIC method. • Linezolid zones should be read with reflected light. • Vancomycin disk diffusion testing is not recommended for *S. aureus* or SOSA. • Daptomycin disk diffusion testing is not recommended.

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Table B1. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal QC[a]	Comments/Modifications
***Enterococcus* spp.**	MHA	Colony suspension in broth or saline, or broth culture method	35°C ± 2°C; ambient air	16–18 hours; 24 hours for vancomycin	*S. aureus* ATCC® 25923	• Vancomycin zones must be examined carefully using transmitted light; the presence of a haze or any growth within the zone of inhibition indicates resistance. Organisms with intermediate zones should be tested by an MIC method as described in CLSI M07.[2] • Daptomycin disk diffusion testing is not recommended.

Abbreviations: ATCC®, American Type Culture Collection; MHA, Mueller-Hinton agar; MIC, minimal inhibitory concentration; MRS, methicillin (oxacillin)-resistant staphylococci; QC, quality control; SOSA, staphylococci other than *Staphylococcus aureus*.

[a] See specific CLSI M100[1] Tables 3 for additional QC recommendations for screening and confirmatory tests.

[b] ATCC® is a registered trademark of the American Type Culture Collection.

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Table B2. Conditions for Disk Diffusion Antimicrobial Susceptibility Tests for Fastidious Organisms (refer to CLSI M100[1] Tables 2)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal QC[a]	Comments/Modifications
Haemophilus influenzae* and *Haemophilus parainfluenzae	HTM (for all agents when testing *H. influenzae* or *H. parainfluenzae*) or MH-F agar (for selected agents when testing *H. influenzae*)	Colony suspension in broth or saline prepared from an overnight (preferably 20- to 24-hour) chocolate agar plate[b]	35°C ± 2°C; 5% CO_2	16–18 hours	• *H. influenzae* ATCC®[c] 49247 (for HTM or MH-F agar/broth) and/or *H. influenzae* ATCC® 49766 (for HTM only) • For β-lactam combination agents, refer to CLSI M100[1] Tables 4.	• Test a maximum of 9 disks on a 150-mm plate and 4 disks on a 100-mm plate. • Use either *H. influenzae* ATCC® 49247 or *H. influenzae* ATCC® 49766 or both of these strains, based on the antimicrobial agents to be tested. Neither strain has QC ranges for all agents that might be tested against *H. influenzae* or *H. parainfluenzae*.
Neisseria gonorrhoeae	GC agar base with 1% defined growth supplement	Colony suspension in MHB or 0.9% phosphate-buffered saline, pH 7, prepared from an overnight (20- to 24-hour) chocolate agar plate incubated in 5% CO_2	36°C ± 1°C (do not exceed 37°C); 5% CO_2	20–24 hours	*N. gonorrhoeae* ATCC® 49226	Test a maximum of 9 disks on a 150-mm plate and 4 disks on a 100-mm plate. For some antimicrobial agents (eg, fluoroquinolones or cephalosporins) only 2–3 disks may be tested per plate.
Streptococcus pneumoniae	MHA with 5% sheep blood or MH-F agar	Colony suspension in broth or saline using colonies from an overnight (18- to 20-hour) sheep blood agar plate	35°C ± 2°C; 5% CO_2	20–24 hours	*S. pneumoniae* ATCC® 49619	• Test a maximum of 9 disks on a 150-mm plate and 4 disks on a 100-mm plate. • Measure the zone of growth inhibition, not the zone of inhibition of hemolysis. • Deterioration of oxacillin disk content is best assessed with *S. aureus* ATCC® 25923, with an acceptable range of 18–24 mm on unsupplemented MHA.

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Table B2. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal QC[a]	Comments/Modifications
***Streptococcus* spp.**	MHA with 5% sheep blood	Colony suspension in broth or saline using colonies from an overnight (18- to 20-hour) sheep blood agar plate	35°C ± 2°C; 5% CO_2	20–24 hours	*S. pneumoniae* ATCC® 49619	• Test a maximum of 9 disks on a 150-mm plate and 4 disks on a 100-mm plate. • Measure the zone of growth inhibition, not the zone of inhibition of hemolysis.
Neisseria meningitidis	MHA with 5% sheep blood[d]	Colony suspension in broth or saline prepared from a 20- to 24-hour chocolate agar plate incubated in 5% CO_2	35°C ± 2°C; 5% CO_2	20–24 hours	• *S. pneumoniae* ATCC® 49619 (5% CO_2) • *E. coli* ATCC® 25922 (ambient air or 5% CO_2; for ciprofloxacin, nalidixic acid, minocycline, and sulfisoxazole)	• Test a maximum of 5 disks on a 150-mm plate and 2 disks on a 100-mm plate. • **Caution:** Perform all testing in a BSC.

Abbreviations: ATCC®, American Type Culture Collection; BSC, biological safety cabinet; CFU, colony-forming unit(s); HTM, *Haemophilus* test medium; MHA, Mueller-Hinton agar; MHB, Mueller-Hinton broth; MH-F, Mueller-Hinton fastidious; pH, negative logarithm of hydrogen ion concentration; QC, quality control.

[a] See specific CLSI M100[1] Tables 3 for additional QC recommendations for screening and confirmatory tests.

[b] This suspension will contain approximately 1 to 4 × 10^8 CFU/mL. Care should be used in preparing this suspension, because higher inoculum concentrations may lead to false-resistant results with some β-lactam antimicrobial agents, particularly when β-lactamase–producing strains of *H. influenzae* are tested.

[c] ATCC® is a registered trademark of the American Type Culture Collection.

[d] Enriched chocolate agar is not recommended for susceptibility testing of *N. meningitidis*.

References for Appendix B

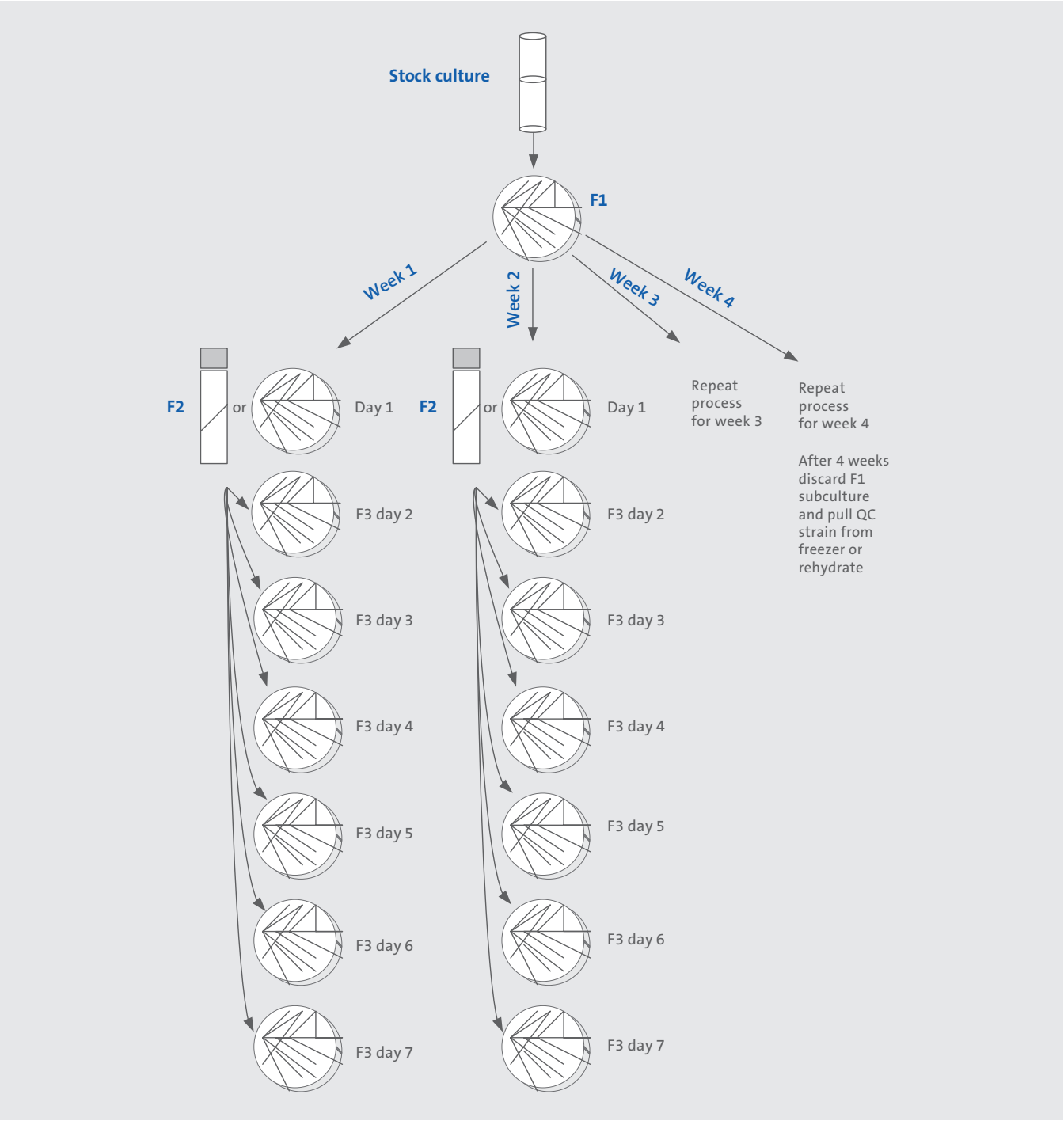
1 CLSI. *Performance Standards for Antimicrobial Susceptibility Testing.* 34th ed. CLSI supplement M100. Clinical and Laboratory Standards Institute; 2024.

2 CLSI. *Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically.* 12th ed. CLSI standard M07. Clinical and Laboratory Standards Institute; 2024.

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Appendix C. Quality Control Strain Maintenance

Figure C1 illustrates the workflow for subculturing and using reference strains as described in Subchapter 4.4 of this standard. In Figure C1, "F" indicates the "frozen" or "freeze-dried" state of the stock culture, while "1" indicates the first passage, "2" the second passage, and "3" the third passage from the stock culture.



Abbreviation: QC, quality control.

Figure C1. Workflow for Subculturing and Using Reference Strains

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Appendix C. (Continued)

The steps for maintaining stock cultures are listed in the following table.

Step	Action	Comment
1	Subculture frozen or freeze-dried stock culture to appropriate agar plate. • Tryptic soy or blood agar for nonfastidious QC strains • Blood agar or enriched chocolate for fastidious QC strains	This subculture is designated the F1 subculture ("F" indicates the "frozen" or "freeze-dried" state of the stock culture, while "1" indicates the first passage and "2" the second passage from the stock culture).
2	Incubate under the appropriate conditions for the organism.	Store F1 subculture at 2 to 8°C or as appropriate for the organism type for up to 4 weeks.
3	Subculture from F1 onto an agar slant or agar plate to prepare F2 subculture.	Subculture lyophilized or frozen stock cultures twice before they are used to prepare a QC test inoculum suspension.
4	Incubate the F2 subculture under the appropriate conditions for the organism.	Store the F2 subculture at 2 to 8°C or as appropriate for the organism type for up to 1 week.
5	Subculture from F2 onto an agar plate (not to broth or agar slant) to prepare F3 subcultures.	The number of F3 subcultures needed depends on the laboratory's QC testing schedule.
6	Select isolated colonies from fresh F2 or F3 agar plate (eg, overnight incubation) to prepare the QC test inoculum suspension.	If a QC test appears contaminated or if QC results are questionable, it may be necessary to prepare a new F1 subculture from lyophilized or frozen stock culture.
7	Prepare a new F2 subculture (agar slant or agar plate) each week for weeks 2, 3, and 4.	• Prepare new F1 subcultures at least every 2 weeks for: – *Pseudomonas aeruginosa* ATCC®[a] 27853 – *Enterococcus faecalis* ATCC® 51299 – *Streptococcus pneumoniae* ATCC® 49619 • If held longer than 2 weeks, results for *P. aeruginosa* ATCC® 27853 and *E. faecalis* ATCC® 51299 may fall out of range and *S. pneumoniae* ATCC® 49619 may lose viability.
8	After 4 weeks, prepare a new F1 subculture from frozen or freeze-dried stock culture.	

Abbreviations: ATCC®, American Type Culture Collection; QC, quality control.
[a] ATCC® is a registered trademark of the American Type Culture Collection.

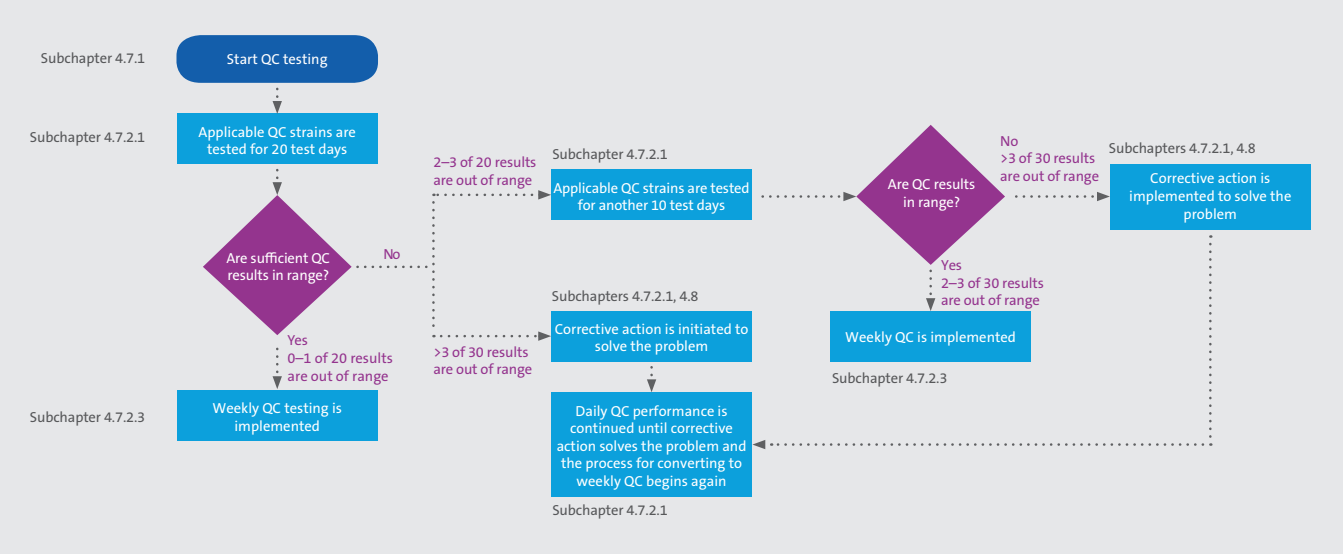
NOTE: The suggestions here apply to daily or weekly QC plans and to routine and supplemental QC strains.

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Appendix D. Quality Control Protocol Flow Charts

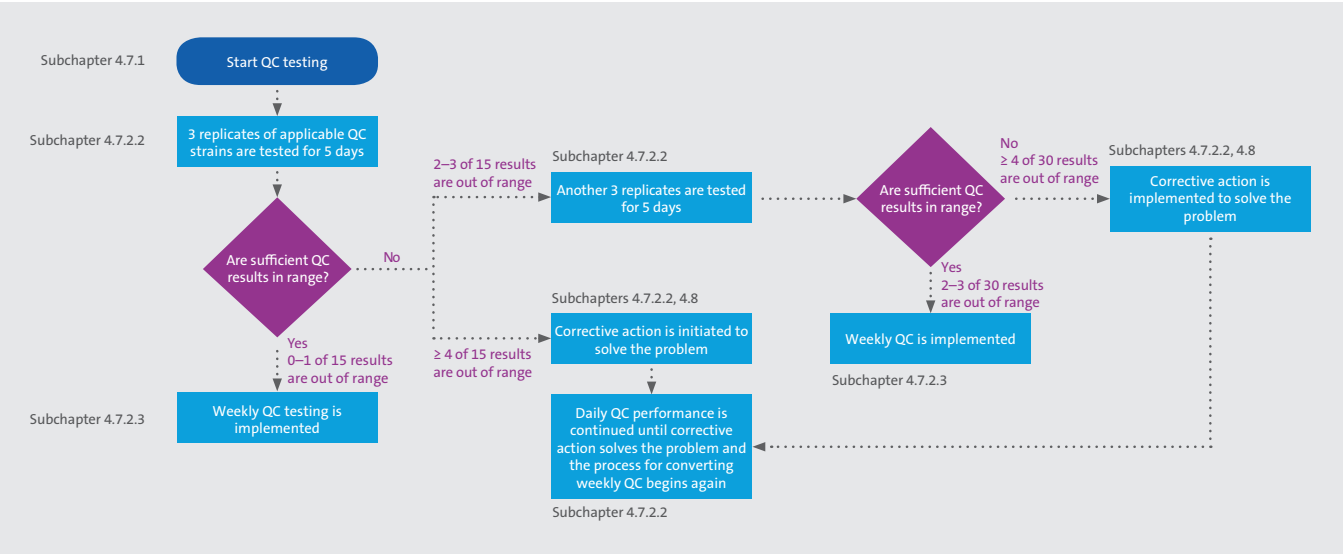
NOTE: All subchapter references are to subchapters in this standard.

D1 Quality Control Protocol: Conversion From Daily to Weekly Testing (20- or 30-Day Plan)



Abbreviation: QC, quality control.

D2 Quality Control Protocol: Conversion From Daily to Weekly Testing (15-Replicate [3- × 5-Day] Plan)

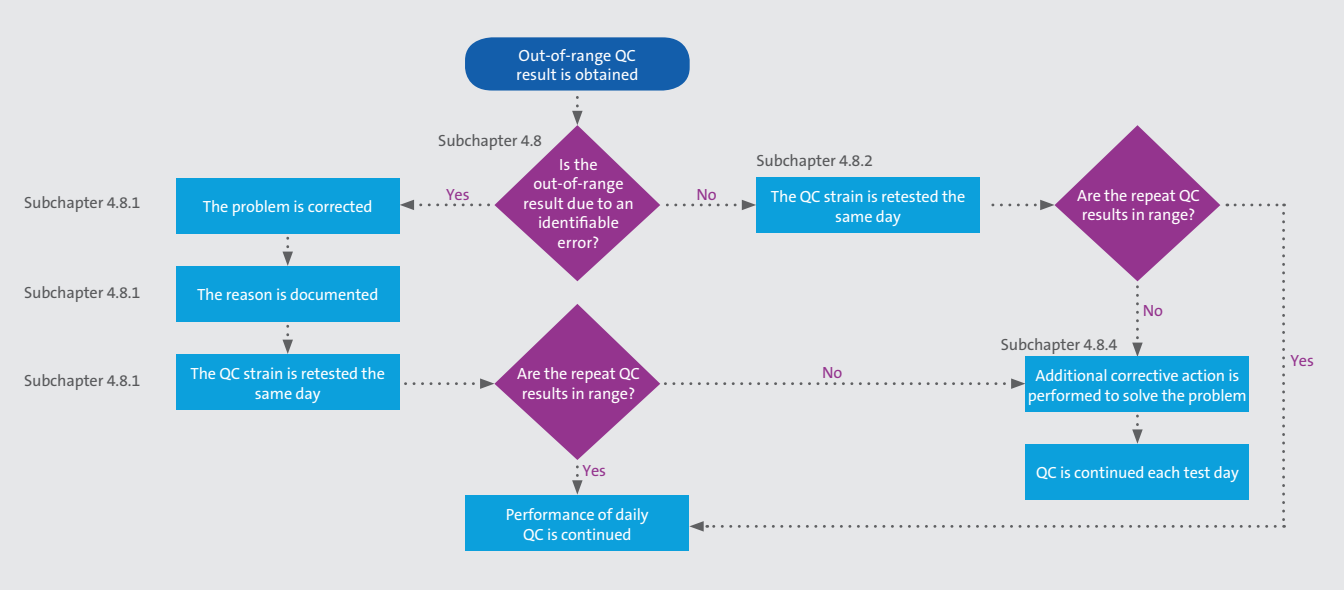


Abbreviation: QC, quality control.

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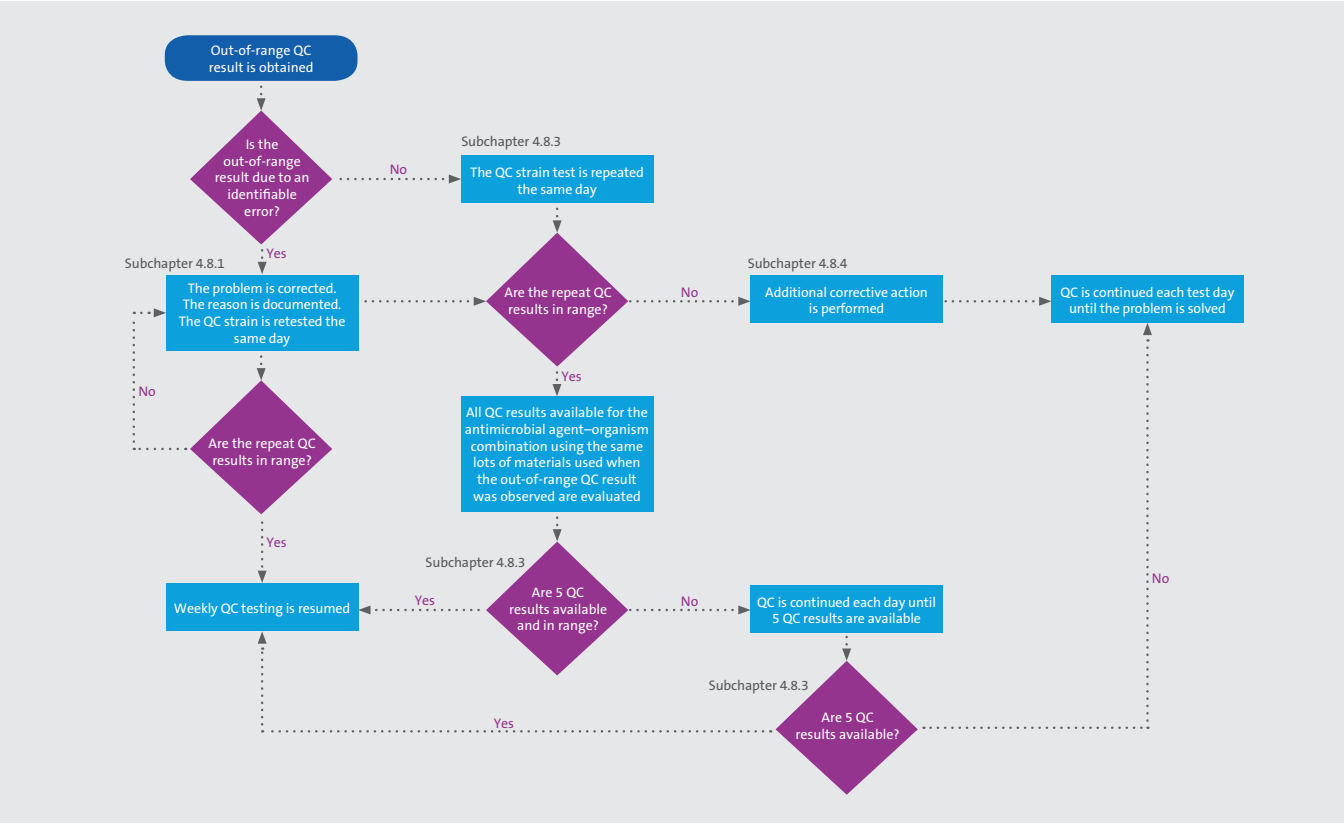
Appendix D. (Continued)

D3 Quality Control Protocol: Daily Quality Control Testing—Corrective Action



Abbreviation: QC, quality control.

D4 Quality Control Protocol: Weekly Quality Control Testing—Corrective Action



Abbreviation: QC, quality control.

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Appendix E. Test for Performing Disk Diffusion Directly From Positive Blood Culture Broth

Test	Direct Disk Diffusion
Test method	Disk diffusion using positive blood culture broth. Specific antimicrobial agent–organism combinations are detailed in CLSI M100[1] Tables 3.
Organism group	Enterobacterales, *Pseudomonas aeruginosa,* and *Acinetobacter* spp.
Medium	MHA
Antimicrobial concentration	Refer to CLSI M100.[1] Standard disk contents for the antimicrobials are detailed in CLSI M100[1] Tables 3.
Inoculum	Positive blood culture broth with gram-negative bacilli, used within 8 hours of flagging positive by the blood culture system
Test procedure	1. Invert blood culture bottle 5–10 times to thoroughly mix. 2. Sterilize the top of the bottle with an alcohol wipe (allow to dry) and insert a 20-gauge venting needle into the blood culture bottle. 3. Dispense 4 drops of blood culture broth onto an MHA plate. As a purity check, use an inoculated blood agar plate streaked for isolation. 4. Spread blood culture broth across the entire surface of the MHA plate using a sterile cotton swab. 5. Repeat this procedure by streaking twice more, rotating the plate approximately 60 degrees each time to ensure an even distribution of inoculum. 6. Leave the lid ajar for 3–5 minutes (ideally) but no more than 15 minutes. 7. Dispense antimicrobial disks onto the surface of the inoculated MHA plate. 8. Press each disk down to ensure complete contact with the agar surface. 9. Invert the plate and place in the incubator within 15 minutes of disks being applied.
Incubation conditions	35°C ± 2°C; ambient air
Incubation length	8–10 hours or 16–18 hours (refer to CLSI M100[1] Tables 3 for antimicrobial agent-specific incubation lengths)
Results	1. Examine the blood agar purity plate to ensure pure growth. 2. Examine the test plate to ensure confluent lawn of growth appropriate to read disk zone tests per this standard. 3. Measure the zone diameters according to routine disk diffusion recommendations in this standard. 4. Interpret results using the zone diameter breakpoints in CLSI M100[1] Tables 3 if the gram-negative bacillus tested is confirmed to be an Enterobacterales, *P. aeruginosa*, or *Acinetobacter* spp. If species is identified as another organism, do not interpret or report results. 5. Report only the interpretative category and not the measured zone size.

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Appendix E. (Continued)

Test	Direct Disk Diffusion
Additional testing and reporting	• If there is an inconsistent growth pattern on the plate (eg, mixed inoculum, nonconfluent growth, growth is too faint to read), do not interpret or report results from the direct disk diffusion test, and perform standard susceptibility testing from pure colony growth. • Antimicrobial agents to which the organism is intrinsically resistant (see CLSI M100[1] appendixes) should be reported as resistant, regardless of measured zone size. • If 2 zones of growth inhibition are observed, measure the inner zone diameter. In case of colonies present within zones, or presence of both inner and outer zones, check the purity plate and, if pure, record the inner zone diameter.
QC recommendations	• Perform QC according to the standard disk diffusion QC procedures per CLSI M02 (eg, daily or weekly). • See CLSI M100[1] Tables 4 for acceptable QC ranges. • *Escherichia coli* ATCC®[a] 25922, *P. aeruginosa* ATCC® 27853, *E. coli* ATCC® 35218 • Refer to CLSI M100[1] Tables 4 to select additional strains for routine QC of β-lactam combination agents, if necessary.

Abbreviations: ATCC®, American Type Culture Collection; MHA, Mueller-Hinton agar; QC, quality control.
[a] ATCC® is a registered trademark of the American Type Culture Collection.

Reference for Appendix E

1 CLSI. *Performance Standards for Antimicrobial Susceptibility Testing.* 34th ed. CLSI supplement M100. Clinical and Laboratory Standards Institute; 2024.

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The Quality Management System Approach

Clinical and Laboratory Standards Institute (CLSI) subscribes to a quality management system (QMS) approach in the development of standards and guidelines that facilitates project management, defines a document structure using a template, and provides a process to identify needed documents. The QMS approach applies a core set of "quality system essentials" (QSEs), basic to any organization, to all operations in any health care service's path of workflow (ie, operational aspects that define how a particular product or service is provided). The QSEs provide the framework for delivery of any type of product or service, serving as a manager's guide. The QSEs are:

- Organization and Leadership
- Customer Focus
- Facilities and Safety Management
- Personnel Management
- Supplier and Inventory Management
- Equipment Management
- Process Management
- Documents and Records Management
- Information Management
- Nonconforming Event Management
- Assessments
- Continual Improvement

The QSEs covered by CLSI M02 and its related CLSI documents are available on the CLSI website: https://clsi.org/qse

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QUICK GUIDE

M02-Ed14-QG

Disk Diffusion Reading Guide

NOTE: Black or dashed lines throughout this guide indicate where the zone of inhibition should be measured.

General Rules for Measuring Zones of Inhibition (Figures 1 to 5)

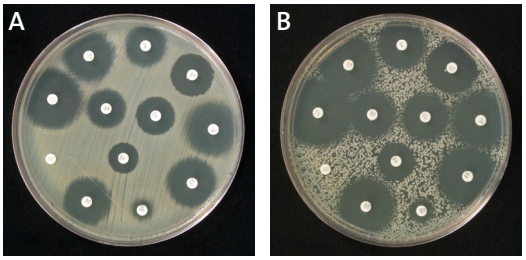


Figure 1. Assessing Growth

- Read plates only when the lawn of growth is confluent (Figure 1A).
- Repeat the test when individual colonies are apparent (Figure 1B).

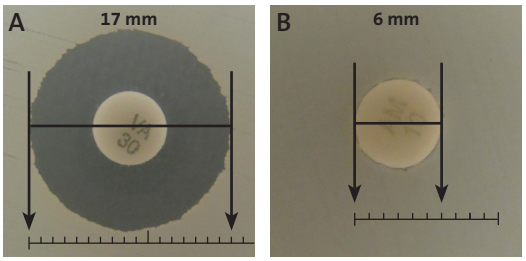


Figure 2. Measuring the Zones of Inhibition

- Measure zones of inhibition to the nearest whole millimeter (mm).
- Zones of complete inhibition include the diameter of the disk and show no obvious, visible growth as judged by the unaided eye (Figure 2A is measured as 17 mm); see Figures 6 to 10 for exceptions.
- Measure growth with no zone of inhibition as 6 mm (Figure 2B).
- Ignore faint growth of tiny colonies that can be detected only with a magnifying lens at the edge of the zone of inhibition.

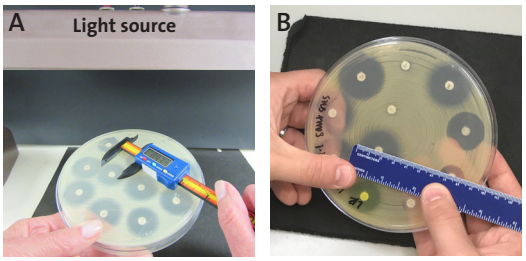


Figure 3. Measuring Zones of Inhibition Using Reflected Light and Translucent Media

- Invert the Petri plate and hold it a few inches above a black background that does not reflect light. Use a light source above the plate to read the zones.
- Measure complete zones of inhibition from the back of the inverted Petri plate (Figures 3A and 3B).

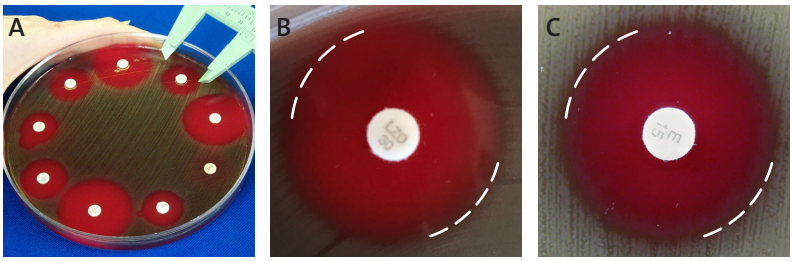


Figure 4. Measuring Zones of Inhibition Using Reflected Light and Opaque Media Supplemented With Blood

- Remove the Petri plate lid and measure zones of inhibition from the upper surface of the agar using a light source above the plate to read the zones.
- Measure the zone of growth inhibition, not the zone of hemolysis. Tilt plate to better differentiate between hemolysis and growth (Figures 4A to 4C).

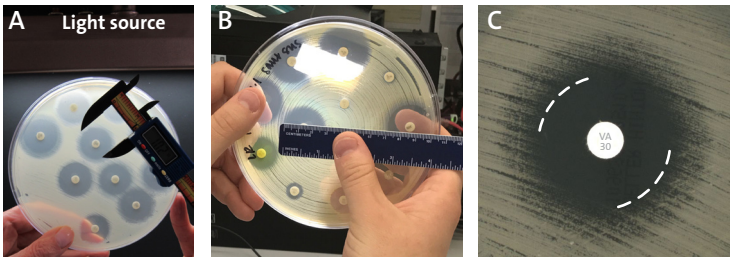


Figure 5. Measuring Zones of Inhibition Using Transmitted Light

- For some tests (eg, *Enterococcus* spp. and vancomycin), measure zones of complete inhibition with the inverted Petri plate placed in front of the light source (transmitted light) as shown in Figures 5A to 5C (see CLSI M100).

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Reference: CLSI M100 | *Performance Standards for Antimicrobial Susceptibility Testing, 34th Edition*

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QUICK GUIDE

M02-Ed14-QG

Disk Diffusion Reading Guide

NOTE: Black or dashed lines throughout this guide indicate where the zone of inhibition should be measured.

Measuring Zones of Inhibition in Special Situations (Figures 6 to 10)

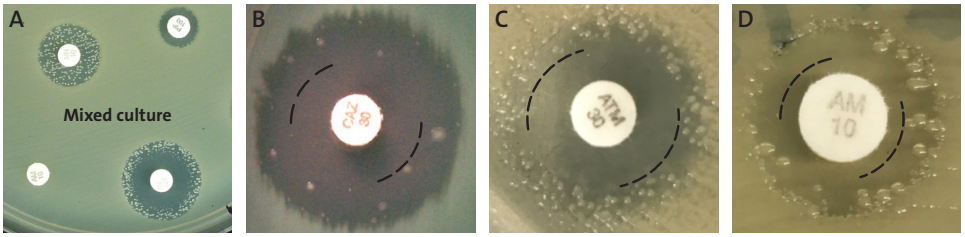


Figure 6. Discrete Colonies Growing Within the Zone

- When discrete colonies grow within a clear zone of inhibition (Figure 6A), repeat the test with a pure culture or subculture of a single colony from the primary culture plate.
- When the discrete colonies continue to grow within the zone of inhibition after repeating the test, measure the colony-free inner zone (Figures 6B to 6D). Measure as 6 mm if there is not a colony-free inner zone.

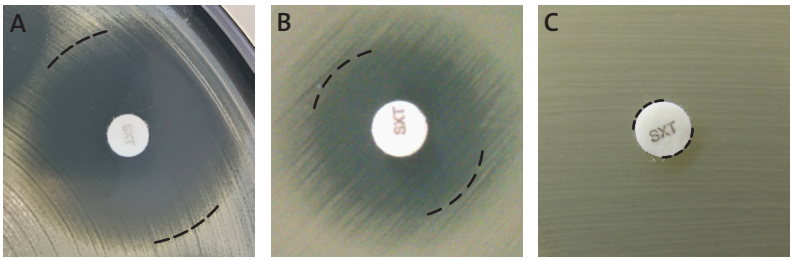


Figure 7. Trimethoprim-Sulfamethoxazole

- When measuring the zone of inhibition for trimethoprim-sulfamethoxazole, disregard slight or hazy growth (20% or less of the lawn of growth) and measure the obvious zone margin as shown in Figures 7A to 7C.

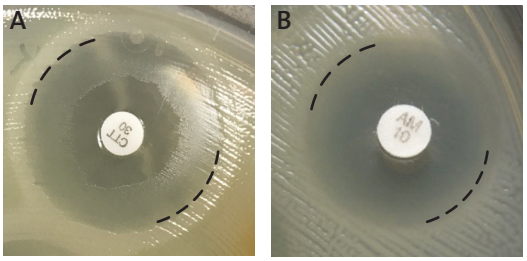


Figure 8. Swarming

- Strains of *Proteus* spp. may swarm into areas of inhibited growth. Ignore the thin veil of swarming growth in an otherwise obvious zone of inhibition as shown in Figures 8A and 8B.

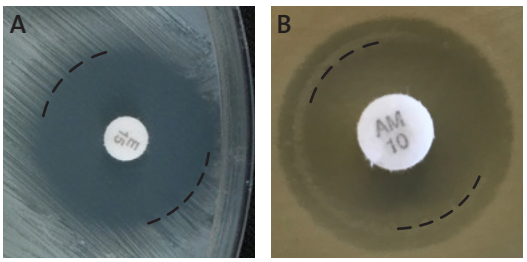


Figure 9. Double Zones

- When double zones are observed, check the growth for purity and repeat the test, if necessary.
- If the culture is pure, measure the inner zone as shown in Figures 9A and 9B.

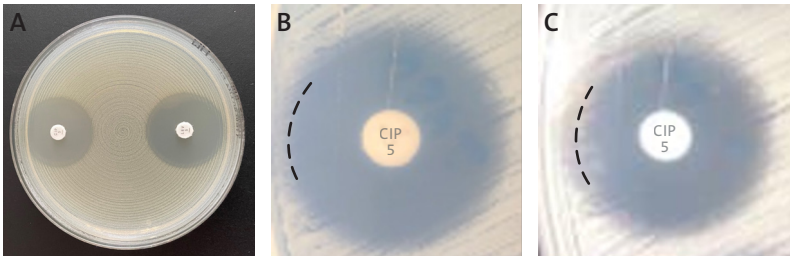


Figure 10. Fuzzy Zones

- Fuzzy zone edges may be seen with some strains of staphylococci.
- It is helpful to tilt the plate to 45 degrees until a clearer zone edge can be observed (Figure 10B).
- Figure 10C is the same plate, but not tilted at 45 degrees, so a clear zone cannot be determined.

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